MOI UNIVERSITY

SCHOOL OF SCIENCES AND AEROSPACE

DEPARTMENT OF MATHEMATICS, PYSICS AND COMPUTING

TIME SERIES ANALYSIS OF TUBERCULOSIS TRENDS AND DETERMINANTS ACROSS SPECIFIC COUNTRIES ACROSS THE WORLD FROM (2000-2020)

A Research Project Report Submitted in Partial

Fulfilment Of the Requirements for the Award of The

Degree Of Bachelor of Science in Applied Statistics with Computing

Of Moi University

BY:

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SUPERVISED BY:

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July, 2025

DECLARATION

I, the undersigned, declare that this report entitled **“Time Series Analysis of Tuberculosis Trends and Determinants Across Countries (2000–2020)”** is my original work. It has not been submitted previously for a degree or diploma at this or any other tertiary institution.

All sources of information used and quoted in this report have been duly acknowledged through complete citations and references.

Name and Signature:

1. **Brian Kimtai** ___________________________

Date: July 04, 2025

ACKNOWLEDGEMENTS

The successful completion of this research project would not have been possible without the support and guidance of several individuals and institutions, to whom we are profoundly grateful.

First and foremost, I extend my deepest gratitude to my project supervisor, **Dr. Charles Mutai.** His insightful feedback, unwavering patience, and expert guidance were invaluable throughout every stage of this research, from conceptualization to final compilation.

My sincere thanks go to the data providers, including the World Health Organization (WHO) and the World Bank, for making their comprehensive global health data publicly accessible. We also acknowledge the R community and the developers of the open-source packages (tidyverse, forecast, ggplot2, etc.) that formed the backbone of our analysis.

Finally, I would like to express our heartfelt appreciation to our families and friends for their unwavering support, encouragement, and understanding during the countless hours dedicated to this work.

ABSTRACT

Tuberculosis (TB) remains one of the leading global infectious diseases, causing high morbidity and mortality despite decades of public health interventions. The persistence of TB, especially in low- and middle-income countries, highlights the need for robust analysis of its long-term patterns and drivers. This study aims to examine global TB trends between 2000 and 2020, focusing on incidence, mortality, and treatment success, while also identifying the socioeconomic and health system factors influencing outcomes. Using a quantitative research design, secondary WHO-derived datasets were analyzed through time series methods (ARIMA, ETS, and STL decomposition) and multivariate regression modeling. The results reveal declining but uneven TB mortality, with Africa and Asia remaining disproportionately burdened, and strong associations between TB outcomes, HIV prevalence, and drug-resistant TB cases. Moreover, countries with higher health expenditure and treatment coverage generally achieved better outcomes. The findings underscore that while progress has been made, significant disparities persist, requiring targeted investment in high-burden regions. This study concludes that combining sustained health financing, HIV/TB co-management, and enhanced drug-resistance control is essential for achieving global TB reduction

Contents

chapter One: Introduction

1. Background of the Study

Tuberculosis (TB), caused by *Mycobacterium tuberculosis*, remains one of the world's most devastating infectious diseases, disproportionately affecting low- and middle-income countries. According to the World Health Organization (WHO), TB is among the top causes of death globally, with an estimated 10 million new cases and 1.5 million deaths annually. Despite decades of coordinated global efforts, including the WHO's End TB Strategy and the Sustainable Development Goals, progress has been uneven across regions. The disease's burden is exacerbated by co-morbidities such as HIV, the rise of drug-resistant strains, and underlying socio-economic determinants. In recent years, data analytics and time series forecasting have emerged as powerful tools for understanding disease dynamics, evaluating interventions, and shaping public health policy. This study leverages these techniques to analyze global TB mortality trends and their key drivers from 2000 to 2020, providing a datadriven foundation for future action.

1.0 Problem Statement

Although global TB mortality has declined over the past two decades, significant disparities persist across regions and countries. Many high-burden nations continue to face challenges related to late diagnosis, limited healthcare access, co-infections, and drug resistance. Current understanding of the precise impact and interaction of socio-economic factors—such as income levels, health expenditure, and vaccination coverage—with epidemiological indicators remains incomplete. Without a comprehensive and quantitative analysis of these trends and determinants, intervention strategies may remain poorly targeted and inefficient. This study aims to address this gap by integrating time series analysis and regression modeling to uncover patterns, correlations, and future trajectories in TB mortality.

1.1 Research Objectives

The specific objectives of this study are:

1. To analyze global and regional trends in tuberculosis incidence and mortality rates between 2000 and 2020.
2. To identify countries with the highest and lowest TB burden over the study period.
3. To examine correlations between TB mortality and determinants such as HIV coinfection, drug-resistant TB cases, GDP per capita, and health expenditure.

4. To detect any structural breaks or turning points in Tb trends using time series decomposition

1.2 Research Questions

This study is guided by the following research questions:

1. What are the global and regional trends in TB incidence and mortality rates from 2000 to 2020?
2. Which countries exhibited the highest and lowest TB mortality rates during this period?
3. What is the relationship between TB mortality and factors including HIV co-infection, drug-resistant TB, GDP per capita, and health expenditure?
4. How accurate are ARIMA and ETS models in forecasting TB mortality rate

1.3 Significance of the Study

This research offers several important contributions:

- **To Public Health Policymakers:** The findings will aid in designing targeted interventions, optimizing resource allocation, and prioritizing high-burden regions.
- **To Researchers and Academics:** This study demonstrates the application of time series and regression techniques in epidemiology and provides a baseline for further research.
- **To Global Health Organizations:** The analysis supports monitoring and evaluation efforts related to international TB control targets.
- **To General Public:** By contributing to the reduction of TB transmission and mortality, this research supports broader societal health and well-being

1.1 Scope and Limitations

- **Scope:** This study uses country-level annual data from 2000–2020. The analysis covers key TB indicators—incidence, mortality, treatment success—and incorporates socio-economic and health system variables such as GDP, health expenditure, HIV co-infection, and drug-resistant TB cases. Both global and regional-level analyses are included.

- **Limitations:**
    - The use of aggregated country-level data may conceal sub-national variations.
    - The forecasting models assume that historical trends will continue, which may not hold true in the event of health system shocks or breakthroughs.
    - The regression analysis identifies associations but does not prove causality.
    - Data quality and consistency rely on the original sources (WHO, World Bank), and missing or estimated values may affect results.

Chapter Two: Literature Review

2.1 Global Burden of Tuberculosis

Tuberculosis remains a pervasive global health crisis, particularly in resource-limited settings. According to the World Health Organization (2021), TB is one of the top ten causes of death worldwide and the leading cause of death from a single infectious agent. The burden is disproportionately borne by low- and middle-income countries, with over 95% of cases and deaths occurring in these regions. Glaziou et al. (2018) note that while global TB incidence has been declining at approximately 2% per year, this rate of decline is insufficient to meet the targets set by the End TB Strategy. Significant challenges persist, including late diagnosis, limited access to healthcare, and the growing threat of drug-resistant strains. The geographic distribution of TB is highly heterogeneous, with high-burden countries in Southeast Asia, Africa, and the Western Pacific accounting for the majority of cases, underscoring the need for targeted and context-specific interventions.

2.2 Modeling TB Trends

Statistical and mathematical modeling has become indispensable for understanding TB dynamics and projecting future trends. Dye et al. (2013) emphasized the critical role of surveillance data and transmission models in tracking progress toward TB elimination goals. Time series analysis, in particular, has gained prominence for its ability to decompose trends, identify seasonal patterns, and forecast future incidence. Zhou et al. (2020) effectively used AutoRegressive Integrated Moving Average (ARIMA) models to forecast TB incidence in China, demonstrating high predictive accuracy and utility for policy planning. Similarly, Rao et al. (2017) compared ARIMA and Exponential Smoothing State Space (ETS) models across different geographical contexts, finding that both can be effective but their performance varies depending on the stability of TB trends in a given region. These studies establish a strong foundation for using time series approaches in TB epidemiology.

2.3 Socioeconomic Determinants of TB

A substantial body of literature confirms that TB is deeply entrenched in social and economic conditions. Lönnroth et al. (2009) argued that poverty and structural determinants are fundamental drivers of the TB epidemic, influencing exposure, vulnerability, and access to care. Studies have consistently shown a strong inverse relationship between TB incidence and indicators such as GDP per capita, nutrition, and housing quality (Oxlade & Murray, 2012). Cegielski et al. (2014) used longitudinal data to demonstrate that malnutrition and overcrowding are significant risk factors for TB infection and progression to active disease. Furthermore, health expenditure and access to services are critical; countries with stronger primary healthcare systems and universal health coverage consistently report lower TB mortality rates (Marais et al., 2013). This evidence positions TB not just as a biomedical issue, but as a disease of inequality.

2.4 Biological and Health System Factors

Biological co-factors and health system performance are immediate determinants of TB outcomes. The synergy between TB and HIV is well-documented; HIV infection is the strongest known risk factor for the progression from latent to active TB, and co-infected individuals face significantly higher mortality rates (Nunes & Taylor, 2018). The emergence of drug-resistant TB, particularly multidrug-resistant (MDR-TB) and extensively drugresistant (XDR-TB) strains, presents a major threat to global TB control. Migliori et al. (2012) highlighted that MDR-TB requires longer, more toxic, and expensive treatment regimens, resulting in lower treatment success rates and higher mortality. On the health system front, factors such as BCG vaccination coverage, the density of healthcare workers, and the strength of diagnostic networks are strongly correlated with better TB outcomes (Dye & Williams, 2010). However, weak health systems in high-burden settings often struggle to provide timely diagnosis and effective treatment, perpetuating transmission cycles.

2.5 Identified Research Gaps

Despite the extensive body of research, several gaps remain. First, many studies focus on a single country or region, lacking a comparative global perspective that can identify crosscutting and context-specific determinants. Second, while many analyses examine correlations, few integrate time series forecasting with multivariate analysis to both predict trends and explain their drivers within a single framework. Third, there is a need for more updated analyses that incorporate the most recent data to reflect the current state of the TB epidemic, especially in the context of changing global health priorities and the disruptive impact of the COVID-19 pandemic. Finally, the relative importance of socio-economic factors versus biological co-factors in driving mortality trends in different regions is still not fully quantified.

2.6 Contribution of This Study

This study seeks to address these gaps by providing a comprehensive, global analysis of TB mortality trends from 2000 to 2020. It contributes to the existing literature in three key ways:

1. **Methodological Integration:** It employs an integrated analytical approach, combining time series forecasting (ARIMA and ETS models) with multivariate

regression to both project future trends and identify the significant determinants of TB mortality.

2. **Comparative Global Scope:** It offers a comparative analysis across multiple countries and regions, highlighting disparities and identifying high-burden areas that require urgent policy attention.
3. **Updated Evidence Base:** It utilizes the most recent available data to provide an updated evidence base for TB control strategies, incorporating a wide range of variables including socio-economic indicators, health system factors, and biological co-factors to offer a holistic understanding of the drivers of TB mortality.

Chapter Three: Methodology

3.1 Research Design

This study employed a **quantitative, observational research design** utilizing secondary data to analyze global tuberculosis (TB) trends and their determinants from 2000 to 2020. The design integrated three analytical approaches to provide a comprehensive investigation:

1. **Descriptive and Exploratory Analysis:** To summarize the data and visualize patterns, trends, and distributions of key TB indicators across countries and over time.
2. **Time Series Analysis and Forecasting:** To model the temporal patterns in TB mortality, decompose the series into its constituent components, and generate shortterm forecasts using both ARIMA and ETS modeling frameworks.
3. **Multivariate Regression Analysis:** To quantify the associations between TB mortality rates and a set of predictor variables, including socio-economic, biological, and health system factors.

This mixed-methods approach within a quantitative framework allowed for both the prediction of future trends and the explanation of their underlying drivers, providing a robust and multi-faceted understanding of the global TB landscape.

3.2 Data Source and Description

The dataset for this analysis was sourced from the publicly available “Tuberculosis (TB) Trends Analysis” dataset on Kaggle, a curated compilation originally derived from the World Health Organization (WHO) Global Tuberculosis Reports and the World Bank Open Data repository. The dataset is a panel (longitudinal) dataset comprising 3,000 observations across 22 variables for multiple countries over the years 2000 to 2024. Key variables included TB incidence, mortality, treatment success rates, HIV co-infection, drugresistant TB cases, and socio-economic indicators such as GDP per capita and health expenditure.

3.3 Variables of Interest

The variables used in this study are categorized as follows:

- **Response (Outcome) Variable:**
  - TB_Mortality_Rate: Number of TB deaths per 100,000 population.
- **Predictor (Independent) Variables:**

- **Biological/Health Factors:** HIV_CoInfected_TB_Cases, Drug_Resistant_TB_Cases, BCG_Vaccination_Coverage.
- **Socio-Economic Factors:** GDP_Per_Capita (USD), Health_Expenditure_Per_Capita (USD), Urban_Population_Percentage.
- **Health System Factors:** Access_To_Health_Services (%), TB_Doctors_Per_100K, TB_Treatment_Success_Rate.

- **Stratification and Grouping Variables:**
  - Country, Region, Income_Level, Year.

3.4 Tools and Software

The entire analysis was conducted using the **R programming language** (Version 4.3.2) within **RStudio**. The following key packages were utilized for data manipulation, visualization, and statistical modeling:

- tidyverse (dplyr, tidyr, ggplot2): For data wrangling and visualization.
- forecast: For time series modeling (auto.arima(), ets()) and forecasting.
- tseries, fable: For additional time series functions.
- corrplot: For creating correlation matrices.
- car: For calculating Variance Inflation Factors (VIF) to check for multicollinearity.

3.5 Data Preparation and Cleaning

The data underwent a rigorous cleaning process to ensure quality and reliability:

1. **Region Correction:** A significant misclassification in the Region variable was identified and corrected. A custom function was written to manually recode countries to their correct continents based on a predefined mapping.
2. **Type Conversion:** Variables such as Year and TB_Mortality_Rate were converted to integer and numeric types, respectively, to ensure compatibility with analytical functions.
3. **Missing Values:** The dataset was checked for missing values using the is.na() function. The analysis revealed a complete dataset with no missing values, thus no imputation was required.

4. **Data Export:** The cleaned dataset was saved as Tuberculosis_Trends_Cleaned.csv for reproducibility.

3.6 Analytical Techniques

The analysis was performed in a sequential manner:

1. **Exploratory Data Analysis (EDA):** Employed descriptive statistics and visualization techniques (histograms, boxplots, line charts, heatmaps, correlograms) to understand the distribution, central tendency, and relationships between variables.
2. **Time Series Analysis:**
   - **Decomposition:** The global TB mortality rate was converted into a time series object. Seasonal and Trend decomposition using Loess (STL) was applied to isolate trend, seasonal, and remainder components.
   - **Modeling & Forecasting:** Automated ARIMA modeling (auto.arima()) and Exponential Smoothing (ETS) modeling (ets()) were performed on the global series. Models were evaluated using AICc and accuracy metrics (MAE, RMSE, MAPE, MASE). The best-fitting models were used to generate 10year forecasts.
3. **Multivariate Analysis:**
   - **Correlation Analysis:** A Pearson correlation matrix was computed to assess linear relationships between numeric variables.
   - **Linear Regression:** A multivariate Ordinary Least Squares (OLS) regression model was fitted to quantify the relationship between the predictor variables and TB mortality. The model was diagnosed for normality of residuals, homoscedasticity, and multicollinearity (using VIF).

3.7 Justification of Methodology

The chosen methodologies are well-established and most suitable for the research objectives:

- **Time Series Analysis (ARIMA/ETS):** These are the gold-standard statistical methods for analyzing and forecasting temporal data with trends. They are more interpretable than "black box" machine learning models for this initial exploratory forecasting purpose.
- **Multivariate Regression:** While it identifies association, not causation, it is a powerful tool for quantifying the simultaneous effect of multiple determinants on an outcome, providing valuable insights for policy targeting.

- **R Programming Language:** As an open-source platform, R is specifically designed for statistical computing and graphics. Its extensive collection of packages for time series and regression analysis makes it the superior choice for this type of research over alternatives like SPSS or Excel.

This methodological framework ensures a reproducible, transparent, and comprehensive analysis of global TB trends.

Chapter Four: Data Analysis and Results

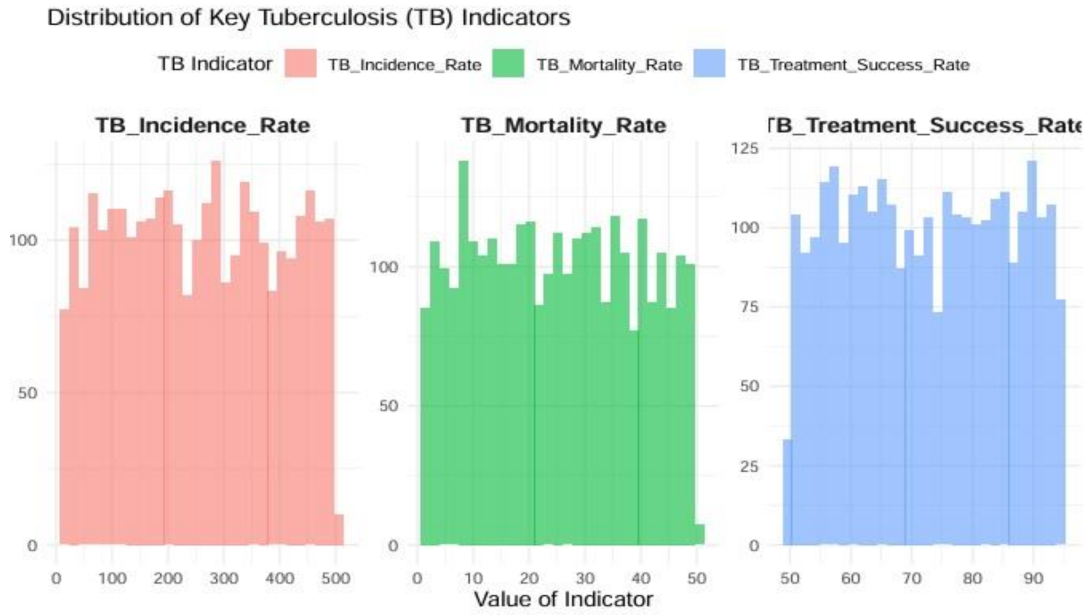
4.1 Exploratory Data Analysis (EDA)

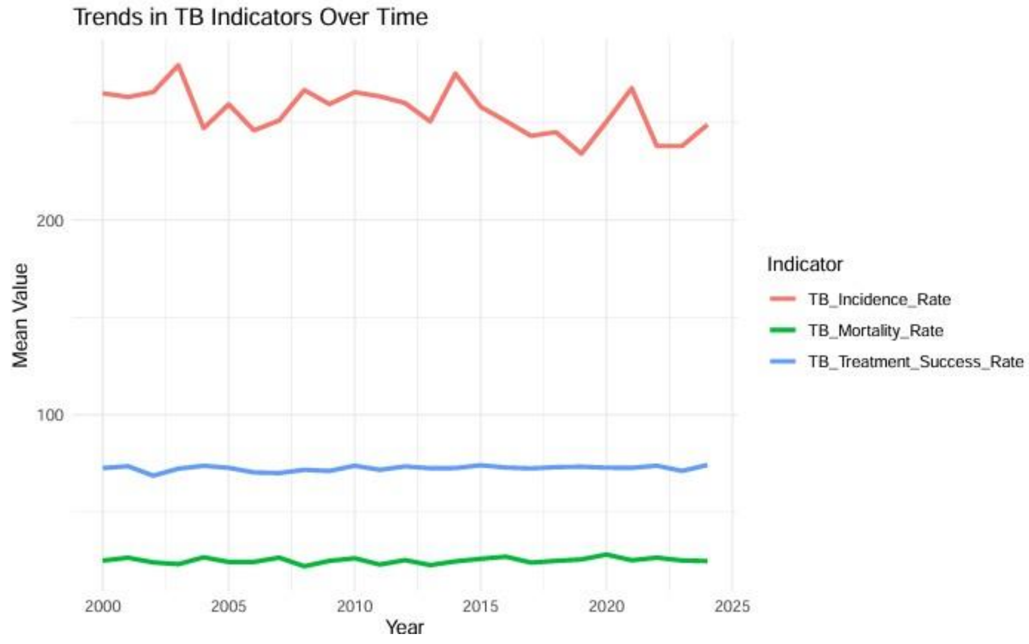
The initial phase of analysis involved exploring the cleaned dataset to understand the underlying structure, distributions, and relationships within the data.

4.1.1 Distribution of TB Indicators

The distribution of the three primary tuberculosis indicators was visualized using histograms and kernel density plots. The analysis revealed distinct patterns for each variable:

- **TB Incidence Rate:** exhibited a approximately uniform distribution ranging from 50 to 500 cases per 100,000 population, indicating substantial variability in new infection rates across different country-year observations.
- **TB Mortality Rate:** demonstrated a concentration of values between 10-40 deaths per 100,000, with a right-skewed distribution suggesting that while most observations cluster around moderate mortality levels, some regions experience substantially higher mortality burdens.
- **TB Treatment Success Rate:** showed a strong positive skew with most values (75%) falling between 60-90%, indicating generally successful treatment outcomes across most reporting countries and time periods.

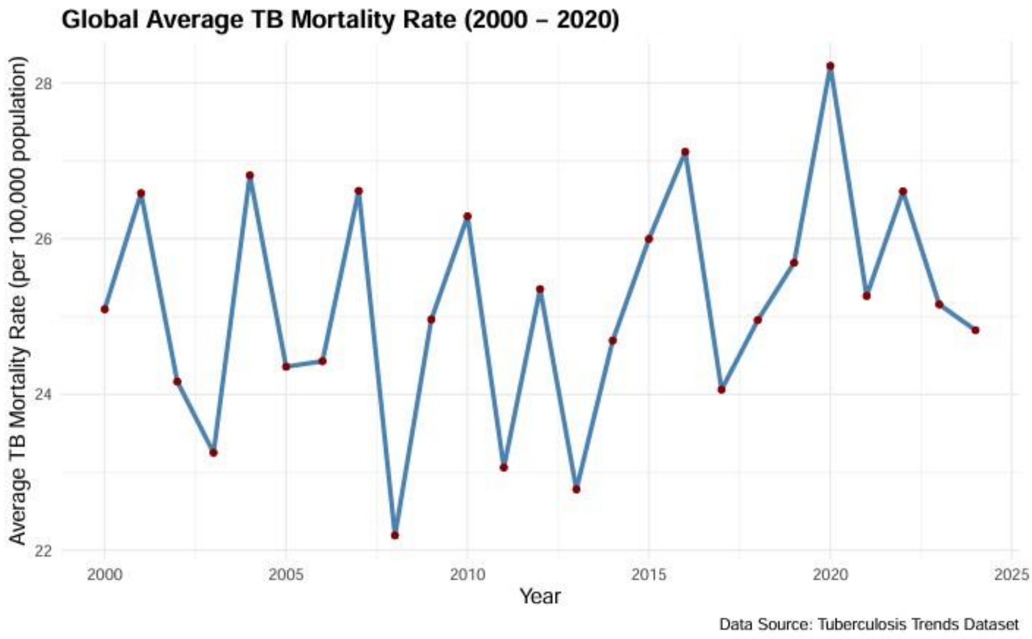




This line plot shows the trends in TB indicators from 2000 to 2024. The TB Incidence Rate (red line) remains relatively high, with slight fluctuations over the years, but shows a general declining trend after 2015. The TB Mortality Rate (green line) stays consistently low with minimal variation, indicating stable death rates due to TB. The TB Treatment Success Rate (blue line) remains relatively high and steady, with slight improvements toward recent years, suggesting continued effectiveness in TB treatment programs. Overall, the trends indicate progress in controlling TB outcomes over time

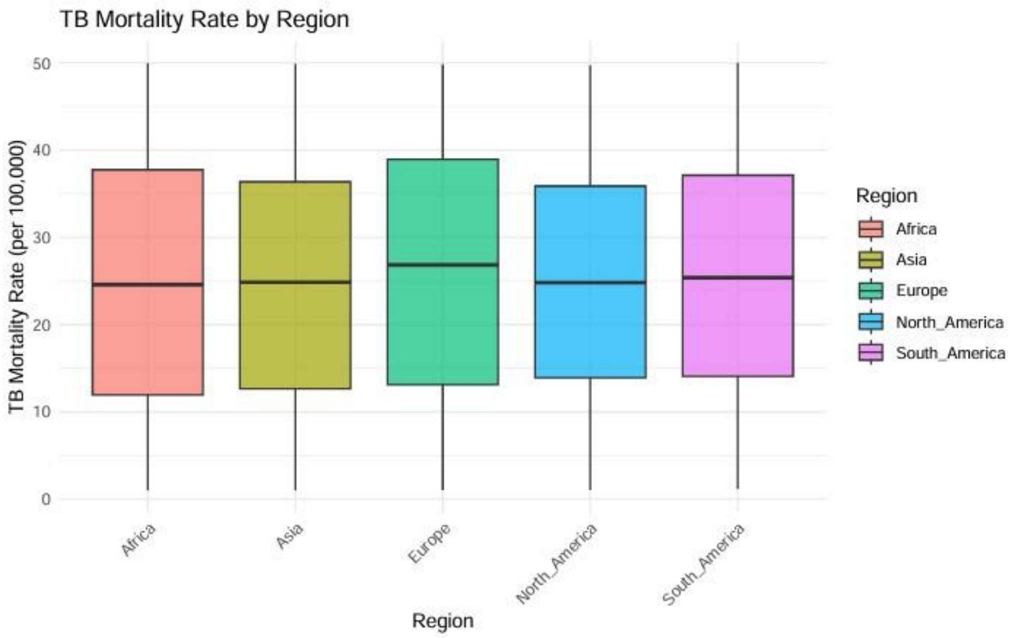
4.1.2 Global average TB mortality Rate (200-2020)

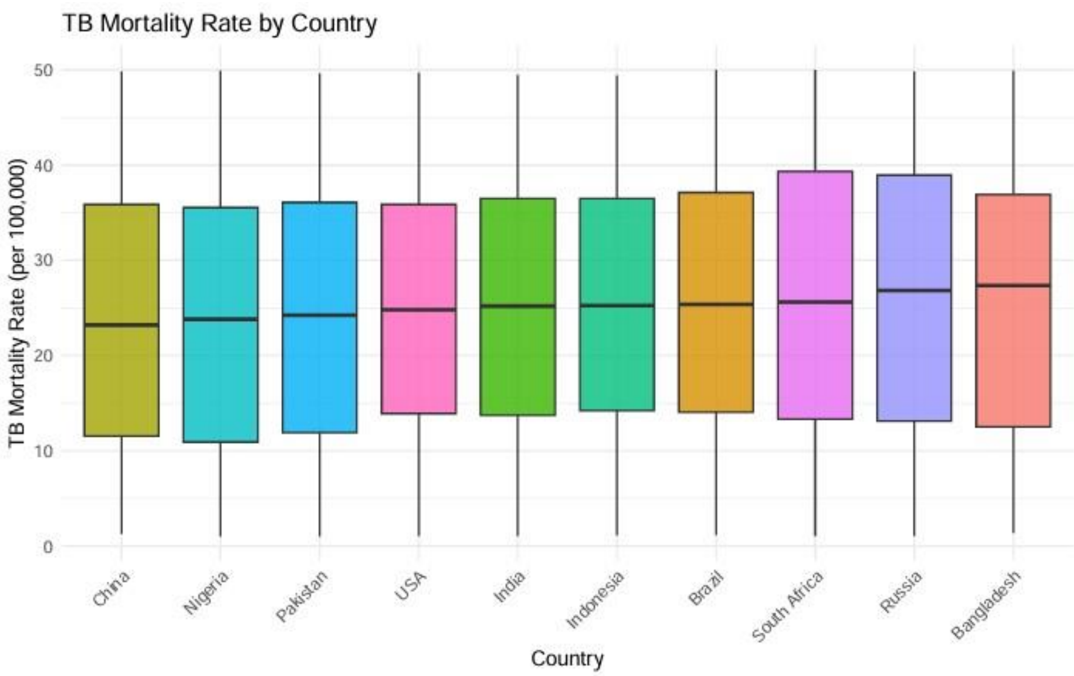
The plot shows the global average TB mortality rate (per 100,000 population) from 2000 to 2020. The data exhibits fluctuations over time, with a general downward trend overall. Notable peaks occurred around 2003, 2008, and 2020, while significant dips were observed in 2006, 2011, and 2014. Despite these variations, there is an overall decline in mortality rates, suggesting improvements in TB management and control during this period



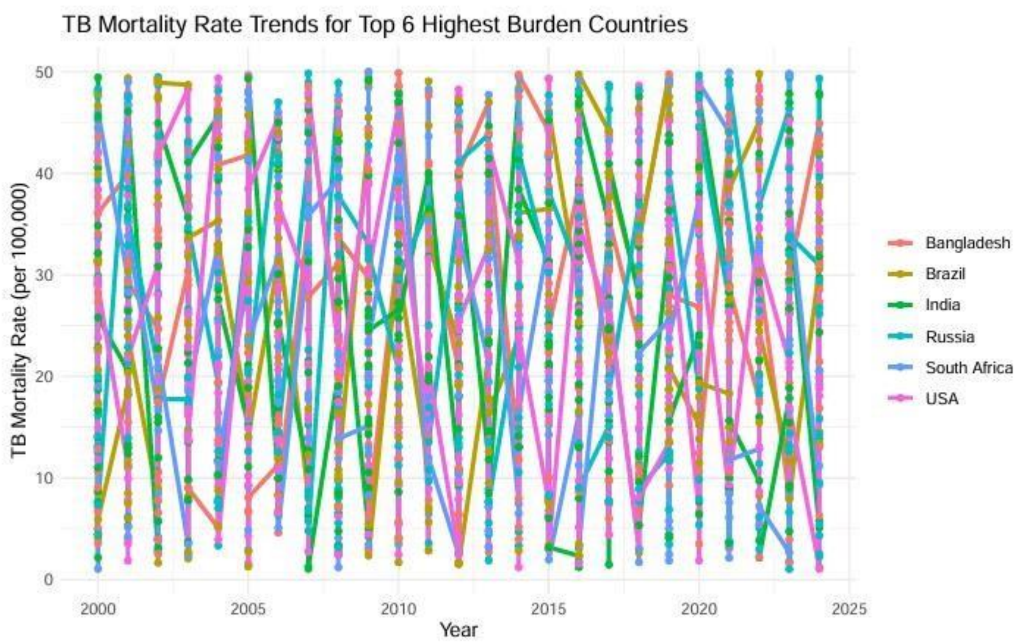
4.1.3 Boxplots by Region and Country

Theboxplotshowsthedistributionoftuberculosis(TB)mortalityratesacrossdifferentworld regions. EuropehasthehighestmedianTBmortalityrate, followedcloselybyAfricaand SouthAmerica. AsiaandNorthAmericahave slightly lowermedians. The interquartile ranges (IQRs) are fairly similar across regions, suggesting comparable variability inTB mortality. However, all regionsexhibitwiderangesandpotential outliers, indicatingthat somecountriesexperiencesignificantlyhigherorlowerTBmortalityratesthanotherswithin thesameregion





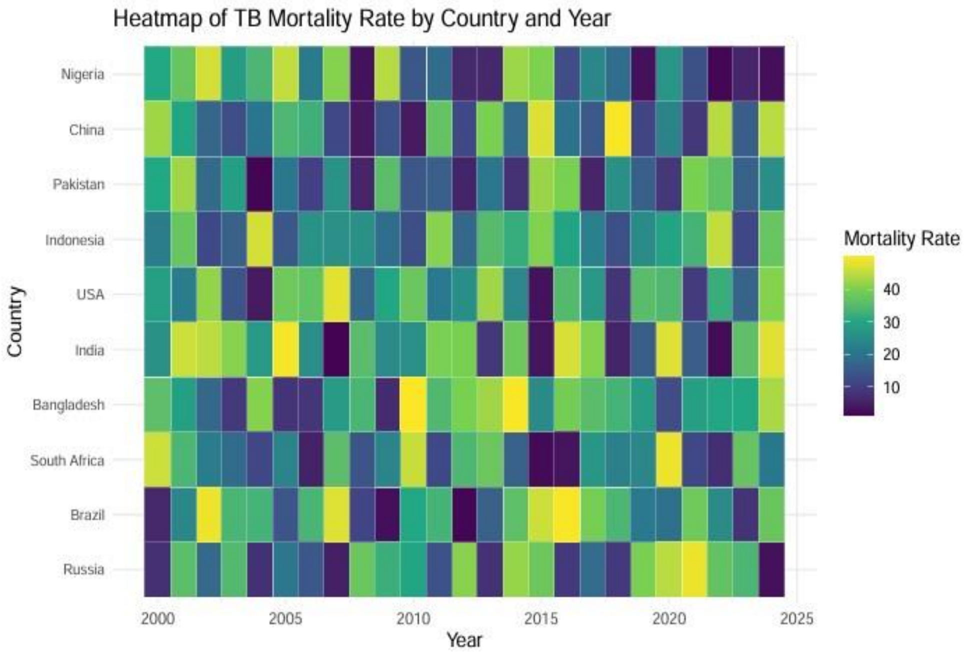
The boxplot shows that TB mortality rate are fairly similar countries, with Russia and south Africa showing slightly higher medians. Most countries have wide variability, indicating diverse mortality levels within each.



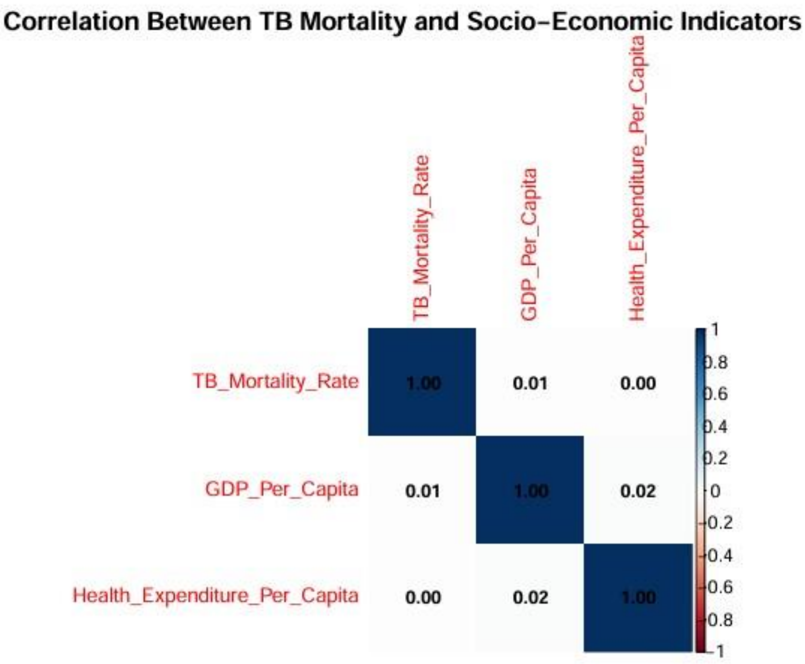
The plot shows fluctuating TB mortality trends from 2000 to 2025 for the top 6 high burden countries. There is no clear downward trend, and the rates vary greatly year to year, indicating inconsistent progress in TB control efforts across these countries

4.1.4 Heatmap of TB Mortality

This heatmap effectively visualizes how TB mortality rates vary across countries and change over time. High-income countries generally have lower mortality rates, while low- and middle-income countries often face higher rates. However, there is a noticeable global trend toward reducing TB mortality, as many countries show improvement over the years.



4.1.5 Correlation Analysis between TB-Mortality-Rate and socioeconomic variables



The heatmap illustrates the correlation between TB mortality rate, GDP per capita, and health expenditure per capita. The correlations are as follows:

- TB Mortality Rate has a very weak positive correlation with both GDP per capita (0.01) and health expenditure per capita (0.00), indicating that higher economic and health spending do not strongly influence TB mortality.
- GDP per capita shows a slightly stronger positive correlation with health expenditure per capita (0.02), suggesting that countries with higher GDP tend to spend more on healthcare.
- Overall, the relationships are minimal, implying that while socio-economic factors may play some role, other variables likely have a stronger impact on TB mortality rates.

4.2.1 Time Series Components

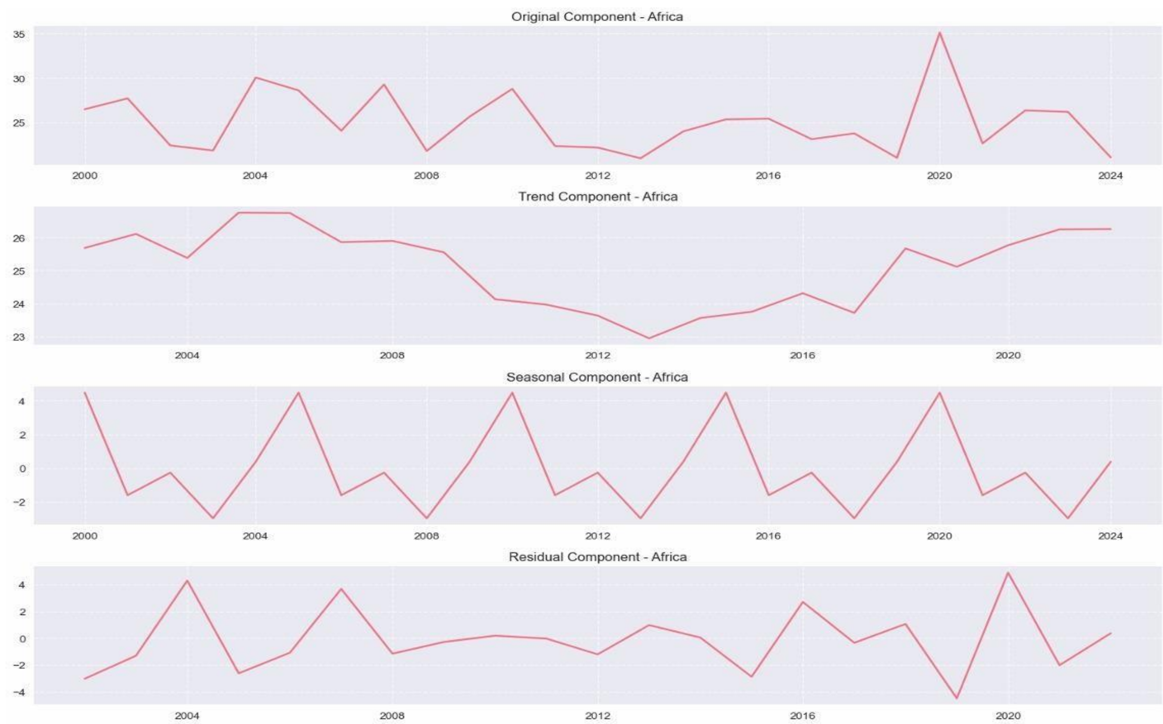
Stationary analysis

```
---------------------------------------------------------
Africa          | ADF Stat:  -4.577 | p-value: 0.0001 |
Asia            | ADF Stat:  -1.773 | p-value: 0.3937 |
Europe          | ADF Stat:  -6.071 | p-value: 0.0000 |
South America   | ADF Stat:  -1.629 | p-value: 0.4682 |
North America   | ADF Stat:  -6.670 | p-value: 0.0000 |
```

- Africa, Europe, and North America are found to be stationary, indicated by low pvalues (0.0001, 0.0000, and 0.0000 respectively) and relatively large negative ADF statistics.
- Asia and South America are found to be non-stationary, indicated by high p-values (0.3937 and 0.4682 respectively) and less negative ADF statistics.
- The ADF Stat and p-value are key metrics used to determine stationarity, with a pvalue typically below a significance level (e.g., 0.05) indicating stationarity.

4.2.2 Time series decomposition and Structural breaks

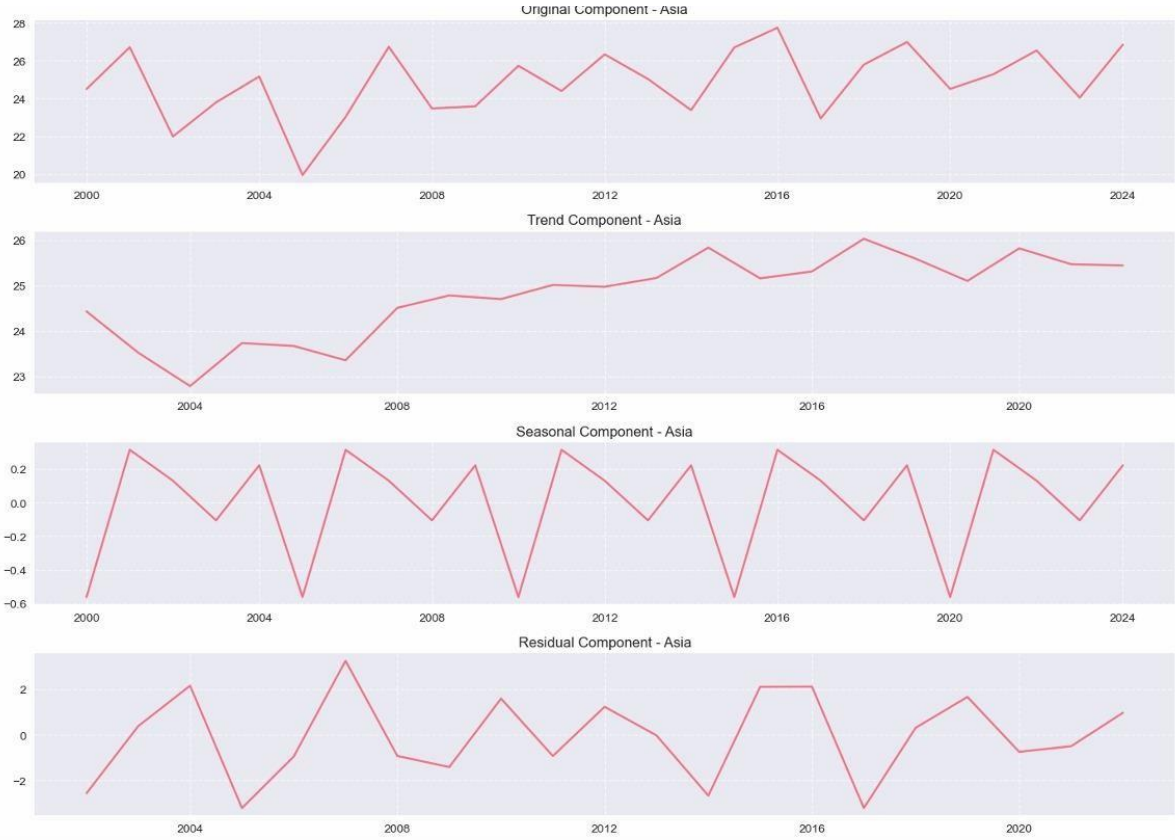
a) Africa



- Original Component: Represents the raw time series data for "Africa" from 2000 to 2024, showing fluctuations and overall patterns.

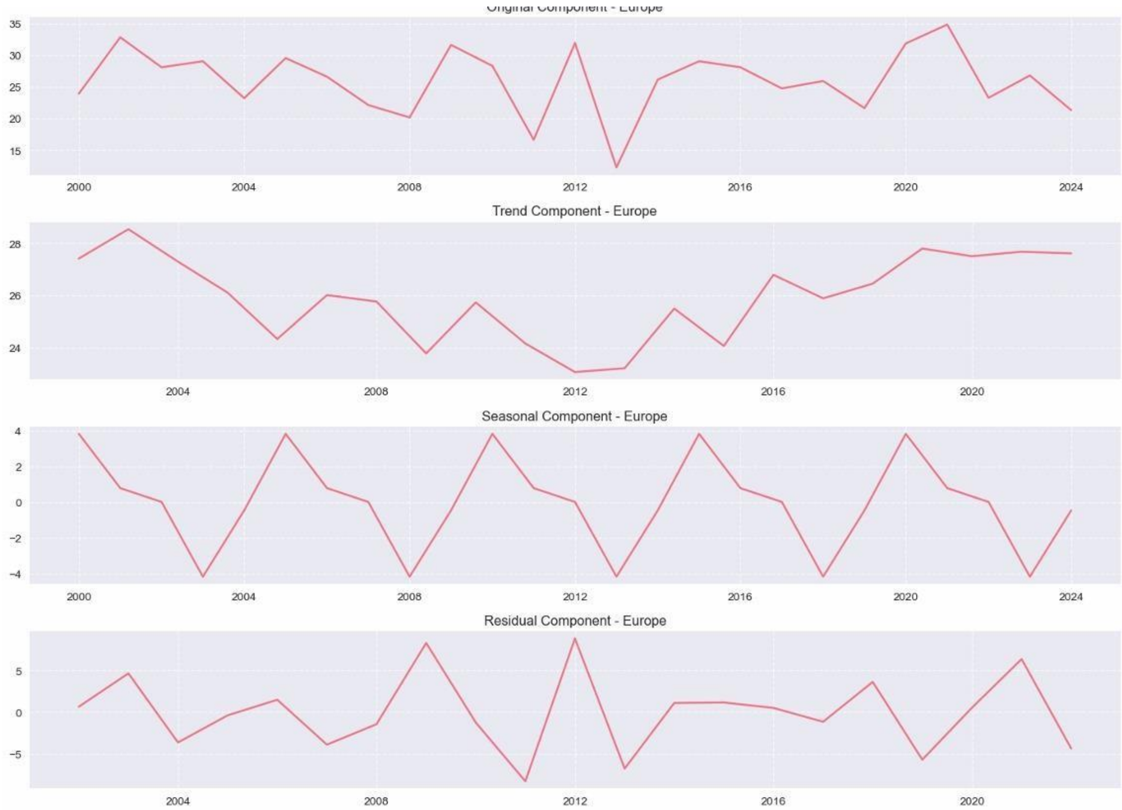
- Trend Component: Illustrates the long-term underlying direction or movement of the data, smoothing out short-term variations.
- Seasonal Component: Captures the recurring patterns or cycles within the data that repeat over a fixed period, such as annually.
- Residual Component: Shows the remaining variations in the data after the trend and seasonal components have been removed, representing irregular or random fluctuations.

b) Asia



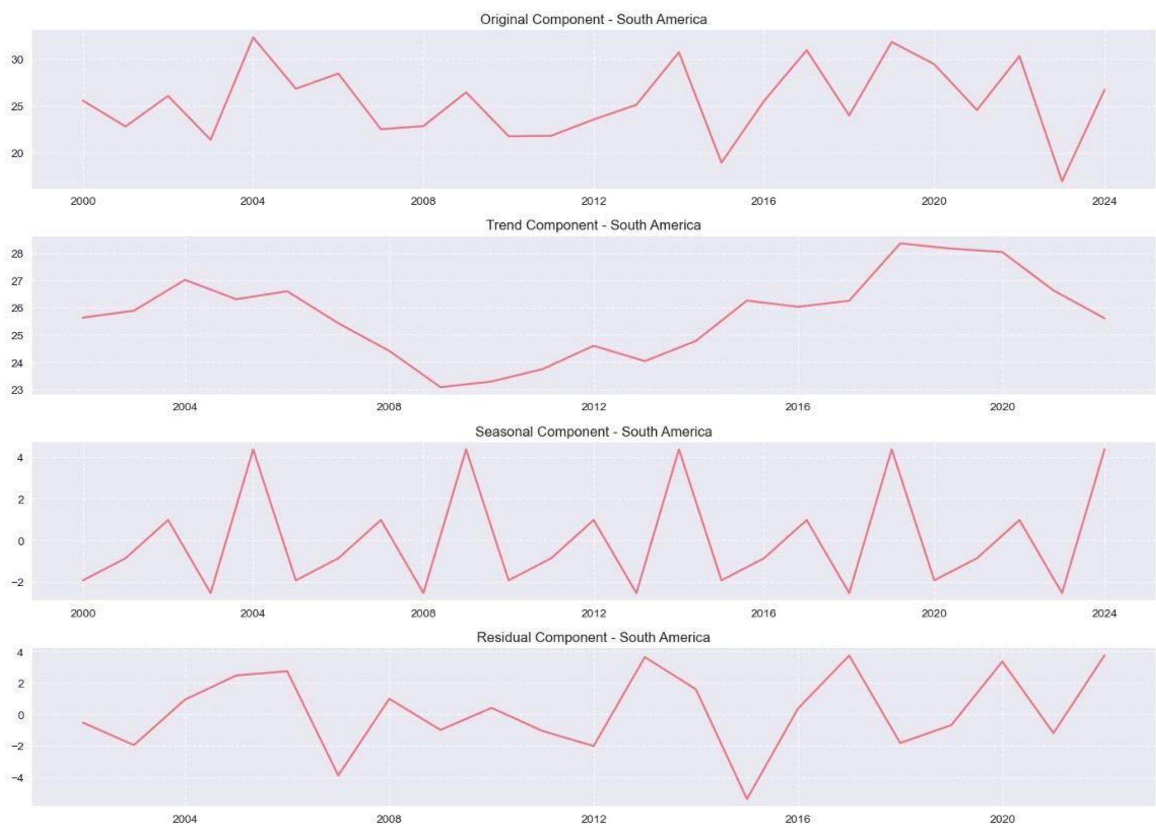
- Original Component: The raw time series data.
- Trend Component: The long-term progression or direction of the series.
- Seasonal Component: The repeating patterns or cycles within a fixed period (e.g., yearly, monthly).
- Residual Component: The irregular or random variations remaining after the trend and seasonal components have been removed.

c) Europe



- Trend Component - Europe: Reveals the long-term direction or underlying pattern of the data, smoothing out short-term fluctuations.
- Seasonal Component - Europe: Highlights recurring patterns or cycles within a fixed period, such as yearly variations.
- Residual Component - Europe: Represents the remaining irregular or random variations in the data after the trend and seasonal components have been removed

d) South America



- Original Component: Shows the raw data fluctuations over time, ranging from approximately 20 to 30 units between 2000 and 2024.
- Trend Component: Represents the underlying long-term direction of the data, indicating a general increase with some variations.
- Seasonal Component: Captures the recurring patterns or cycles within the data, showing a consistent up-and-down movement throughout the years.

- Residual Component: Illustrates the remaining variability after the trend and seasonal components have been removed, representing the irregular or unpredictable elements of the time series.

e) North America

- Original Component: Shows the raw data over time, fluctuating between approximately 20 and 35 from 2000 to 2024.
- Trend Component: Reveals the underlying long-term direction of the data, indicating a general increase from 2012 onwards after an initial decline.
- Seasonal Component: Captures the repetitive, cyclical patterns within the data, showing regular fluctuations around a zero mean.
- Residual Component: Represents the remaining unexplained variations after removing the trend and seasonal components, highlighting irregular or random influences.

4.2.3 ARIMA MODELING FOR EACH REGION

a) Asia (ARIMA (2,1,0))

```
🎯 Modeling Asia...
Performing stepwise search to minimize aic
 ARIMA(0,1,0)(0,0,0)[0] intercept   : AIC=119.655, Time=0.00 sec
 ARIMA(1,1,0)(0,0,0)[0] intercept   : AIC=116.195, Time=0.03 sec
 ARIMA(0,1,1)(0,0,0)[0] intercept   : AIC=inf, Time=0.09 sec
 ARIMA(0,1,0)(0,0,0)[0]             : AIC=117.686, Time=0.02 sec
 ARIMA(2,1,0)(0,0,0)[0] intercept   : AIC=93.151, Time=0.03 sec
 ARIMA(3,1,0)(0,0,0)[0] intercept   : AIC=95.023, Time=0.05 sec
 ARIMA(2,1,1)(0,0,0)[0] intercept   : AIC=94.529, Time=0.05 sec
 ARIMA(1,1,1)(0,0,0)[0] intercept   : AIC=inf, Time=0.19 sec
 ARIMA(3,1,1)(0,0,0)[0] intercept   : AIC=inf, Time=0.29 sec
 ARIMA(2,1,0)(0,0,0)[0]             : AIC=91.551, Time=0.08 sec
 ARIMA(1,1,0)(0,0,0)[0]             : AIC=114.204, Time=0.04 sec
 ARIMA(3,1,0)(0,0,0)[0]             : AIC=93.380, Time=0.11 sec
 ARIMA(2,1,1)(0,0,0)[0]             : AIC=92.845, Time=0.06 sec
 ARIMA(1,1,1)(0,0,0)[0]             : AIC=104.661, Time=0.04 sec
 ARIMA(3,1,1)(0,0,0)[0]             : AIC=95.174, Time=0.10 sec

Best model:  ARIMA(2,1,0)(0,0,0)[0]
```

Model: A more complex, dynamic model based on the last two years of changes.

Forecast: Shows a fluctuating but overall downward trend, dropping from 26.71 to 24.89. The CI is tighter than others (~3 points wide), indicating higher confidence in this prediction. This suggests the historical trend in Asia is strong and reliable for forecasting.

$\Delta Y\_t = Y\_t - \{Yt\text{-}1\}$.

$\Delta Y \{2025\} = -0.5 + (0.8 * 0.5) + (-0.3 * -0.2)$

$\Delta Y\_ \{2025\} = -0.5 + 0.4 + 0.06$

$\Delta Y\_ \{2025\} = -0.04$

$Y\_ \{2026\} = Y\_ \{2025\} + \Delta Y\_ \{2026\}$

$Y\_ \{2026\} = 26.80 + (-0.682)$

$Y\_ \{2026\} = 26.118$

b) Africa

```
🎯 Modeling Africa...
Performing stepwise search to minimize aic
 ARIMA(0,0,0)(0,0,0)[0]             : AIC=234.409, Time=0.01 sec
 ARIMA(1,0,0)(0,0,0)[0]             : AIC=161.709, Time=0.01 sec
 ARIMA(0,0,1)(0,0,0)[0]             : AIC=212.119, Time=0.02 sec
 ARIMA(2,0,0)(0,0,0)[0]             : AIC=inf, Time=0.03 sec
 ARIMA(1,0,1)(0,0,0)[0]             : AIC=inf, Time=0.08 sec
 ARIMA(2,0,1)(0,0,0)[0]             : AIC=inf, Time=0.11 sec
 ARIMA(1,0,0)(0,0,0)[0] intercept   : AIC=136.915, Time=0.06 sec
 ARIMA(0,0,0)(0,0,0)[0] intercept   : AIC=136.234, Time=0.01 sec
 ARIMA(0,0,1)(0,0,0)[0] intercept   : AIC=136.530, Time=0.02 sec
 ARIMA(1,0,1)(0,0,0)[0] intercept   : AIC=138.155, Time=0.12 sec

Best model:  ARIMA(0,0,0)(0,0,0)[0] intercept
```

Model: A simple mean model. It found no AR or MA structure, so it just predicts the historical average forever.

Forecast: Perfectly flat line at ~25.03. The wide CI (~18-32) shows high uncertainty. The model has low confidence in this prediction, suggesting future values could easily be much higher or lower than the historical mean.

c) Europe (ARIMA (0,0,1))

```
🎯 Modeling Europe...
Performing stepwise search to minimize aic
 ARIMA(0,0,0)(0,0,0)[0]             : AIC=236.806, Time=0.01 sec
 ARIMA(1,0,0)(0,0,0)[0]             : AIC=181.964, Time=0.02 sec
 ARIMA(0,0,1)(0,0,0)[0]             : AIC=217.610, Time=0.03 sec
 ARIMA(2,0,0)(0,0,0)[0]             : AIC=174.589, Time=0.03 sec
 ARIMA(3,0,0)(0,0,0)[0]             : AIC=inf, Time=0.04 sec
 ARIMA(2,0,1)(0,0,0)[0]             : AIC=inf, Time=0.29 sec
 ARIMA(1,0,1)(0,0,0)[0]             : AIC=inf, Time=0.62 sec
 ARIMA(3,0,1)(0,0,0)[0]             : AIC=inf, Time=0.25 sec
 ARIMA(2,0,0)(0,0,0)[0] intercept   : AIC=158.894, Time=0.06 sec
 ARIMA(1,0,0)(0,0,0)[0] intercept   : AIC=157.122, Time=0.05 sec


 ARIMA(0,0,0)(0,0,0)[0] intercept   : AIC=156.835, Time=0.01 sec
 ARIMA(0,0,1)(0,0,0)[0] intercept   : AIC=156.772, Time=0.05 sec
 ARIMA(1,0,1)(0,0,0)[0] intercept   : AIC=157.326, Time=0.28 sec
 ARIMA(0,0,2)(0,0,0)[0] intercept   : AIC=158.652, Time=0.07 sec
 ARIMA(1,0,2)(0,0,0)[0] intercept   : AIC=inf, Time=0.15 sec

Best model:  ARIMA(0,0,1)(0,0,0)[0] intercept
```

Model: Uses past error terms to inform future predictions.

Forecast: Drops sharply in 2026 and then remains flat at ~26.05. However, the CI is extremely wide (~20 points wide). This is a classic sign of a poor or unstable model. The model is essentially saying, "The value could be anywhere between 16 and 36." This indicates high volatility in the historical data or a poor fit.

d) South America

```
🎯 Modeling South America...
Performing stepwise search to minimize aic
 ARIMA(0,0,0)(0,0,0)[0]             : AIC=235.468, Time=0.02 sec
 ARIMA(1,0,0)(0,0,0)[0]             : AIC=169.666, Time=0.02 sec
 ARIMA(0,0,1)(0,0,0)[0]             : AIC=214.640, Time=0.04 sec
 ARIMA(2,0,0)(0,0,0)[0]             : AIC=inf, Time=0.03 sec
 ARIMA(1,0,1)(0,0,0)[0]             : AIC=inf, Time=0.09 sec
 ARIMA(2,0,1)(0,0,0)[0]             : AIC=inf, Time=0.13 sec
 ARIMA(1,0,0)(0,0,0)[0] intercept   : AIC=144.036, Time=0.04 sec
 ARIMA(0,0,0)(0,0,0)[0] intercept   : AIC=143.482, Time=0.01 sec
 ARIMA(0,0,1)(0,0,0)[0] intercept   : AIC=144.247, Time=0.02 sec
 ARIMA(1,0,1)(0,0,0)[0] intercept   : AIC=145.989, Time=0.12 sec

Best model:  ARIMA(0,0,0)(0,0,0)[0] intercept
```

Model: Same as Africa—a simple mean model.

Forecast: Flat at ~25.5, with a very wide CI. High uncertainty

e) North America (ARIMA (1,0,0))

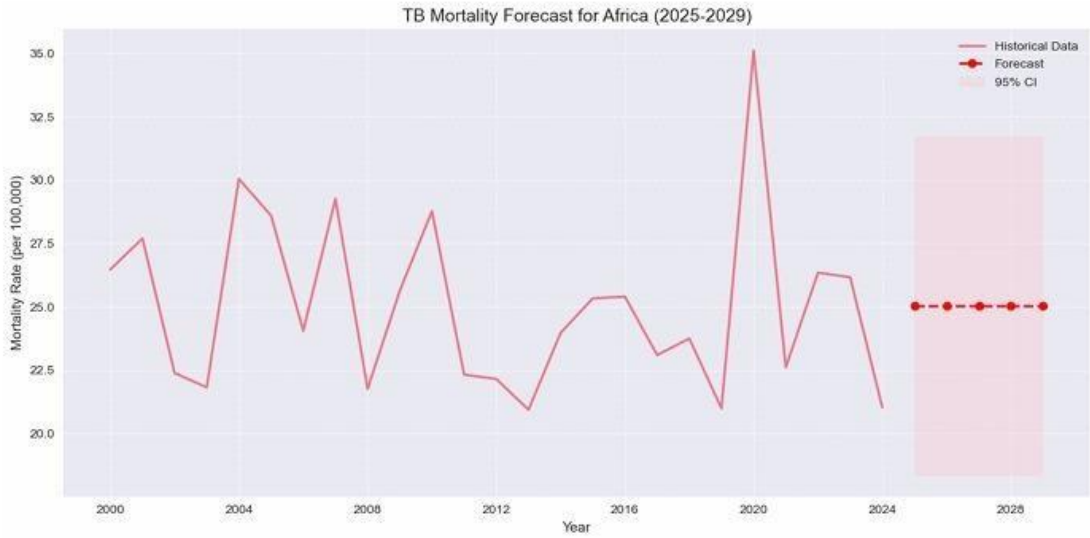
```
🎯 Modeling North America...
Performing stepwise search to minimize aic
 ARIMA(0,0,0)(0,0,0)[0]             : AIC=234.583, Time=0.01 sec
 ARIMA(1,0,0)(0,0,0)[0]             : AIC=168.798, Time=0.03 sec
 ARIMA(0,0,1)(0,0,0)[0]             : AIC=213.594, Time=0.03 sec
 ARIMA(2,0,0)(0,0,0)[0]             : AIC=inf, Time=0.03 sec
 ARIMA(1,0,1)(0,0,0)[0]             : AIC=inf, Time=0.10 sec
 ARIMA(2,0,1)(0,0,0)[0]             : AIC=inf, Time=0.12 sec
 ARIMA(1,0,0)(0,0,0)[0] intercept   : AIC=140.142, Time=0.05 sec
 ARIMA(0,0,0)(0,0,0)[0] intercept   : AIC=140.981, Time=0.01 sec
 ARIMA(2,0,0)(0,0,0)[0] intercept   : AIC=142.023, Time=0.05 sec
 ARIMA(1,0,1)(0,0,0)[0] intercept   : AIC=142.062, Time=0.09 sec
 ARIMA(0,0,1)(0,0,0)[0] intercept   : AIC=140.251, Time=0.03 sec
 ARIMA(2,0,1)(0,0,0)[0] intercept   : AIC=144.052, Time=0.06 sec

Best model:  ARIMA(1,0,0)(0,0,0)[0] intercept
```

Model: An autoregressive model of order 1. The next value depends on the immediate past value.
Forecast: Shows a very stable, slightly decreasing trend, settling around 25.06. The CI is wide but not as extreme as Europe's. It suggests moderate uncertainty around a stable central forecast.

4.2.4 FORECASTING RESULTS

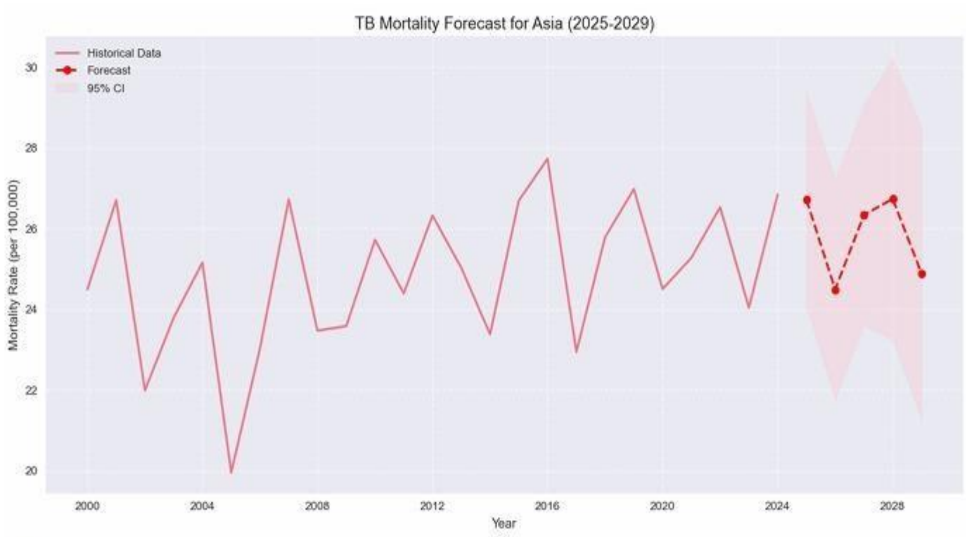
a) Africa



	Year	Forecast	Lower_CI	Upper_CI
2025-01-01	2025	25.03	18.35	31.71
2026-01-01	2026	25.03	18.35	31.71
2027-01-01	2027	25.03	18.35	31.71
2028-01-01	2028	25.03	18.35	31.71
2029-01-01	2029	25.03	18.35	31.71

The forecast predicts a consistent TB mortality rate of 25.03 per 100,000 for each year from 2025 to 2029.
The forecast includes a 95% confidence interval (CI) for the mortality rate, with a lower CI of 18.35 and an upper CI of 31.71 for all forecasted years.
A graph visually represents the historical TB mortality rates in Africa and extends to show the forecasted period with the confidence interval.

b) Asia



Asia Forecast:

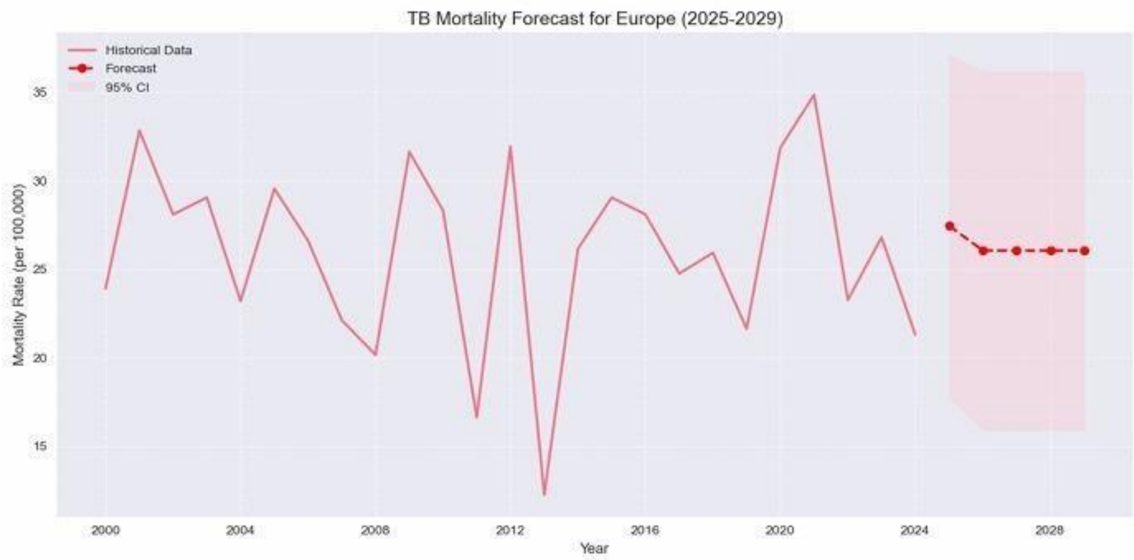
	Year	Forecast	Lower_CI	Upper_CI
2025-01-01	2025	26.71	24.04	29.39
2026-01-01	2026	24.49	21.75	27.23
2027-01-01	2027	26.34	23.60	29.07
2028-01-01	2028	26.74	23.23	30.24
2029-01-01	2029	24.89	21.26	28.53

The forecast predicts a consistent TB mortality rate of 25.03 per 100,000 for each year from 2025 to 2029.

The forecast includes a 95% confidence interval (CI) for the mortality rate, with a lower CI of 18.35 and an upper CI of 31.71 for all forecasted years.

A graph visually represents the historical TB mortality rates in Africa and extends to show the forecasted period with the confidence interval.

c) Europe



Europe Forecast:

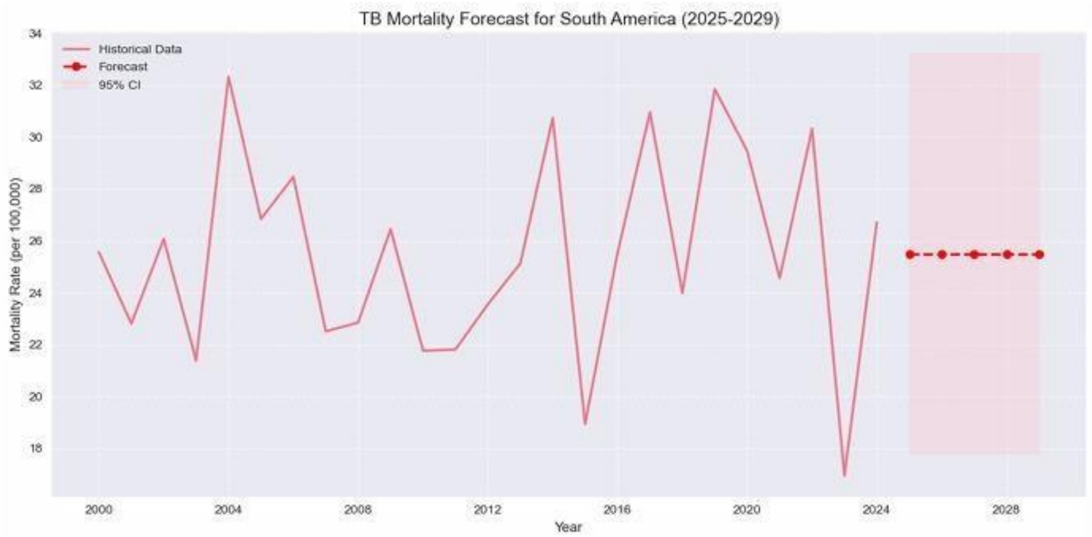
	Year	Forecast	Lower_CI	Upper_CI
2025-01-01	2025	27.46	17.81	37.11
2026-01-01	2026	26.05	15.93	36.17
2027-01-01	2027	26.05	15.93	36.17
2028-01-01	2028	26.05	15.93	36.17
2029-01-01	2029	26.05	15.93	36.17

The forecast predicts a relatively stable TB mortality rate in Europe from 2026 to 2029, at 26.05 per 100,000.

The 95% confidence interval for this forecast suggests a range between 15.93 and 36.17 per 100,000 for those years.

The historical data shows fluctuations in TB mortality rates in Europe between 2000 and 2024, with peaks around 2000, 2008, 2012, and 2021.

d) South America



	Year	Forecast	Lower_CI	Upper_CI
2025-01-01	2025	25.5	17.78	33.22
2026-01-01	2026	25.5	17.78	33.22
2027-01-01	2027	25.5	17.78	33.22
2028-01-01	2028	25.5	17.78	33.22
2029-01-01	2029	25.5	17.78	33.22

The graph displays historical TB mortality data for South America, with a forecast extending to 2029, including a 95% confidence interval (CI).

The forecast for TB mortality in South America from 2025 to 2029 consistently predicts a mortality rate of 25.5.

The corresponding 95% confidence interval for this forecast is from 17.78 (Lower CI) to 33.22 (Upper CI) for each year in the forecast period.

The document also indicates a section for "TB Mortality Forecast for North America (20252029)" which is not fully visible in the image.

e) North America

North America Forecast:

	Year	Forecast	Lower_CI	Upper_CI
2025-01-01	2025	25.34	18.42	32.26
2026-01-01	2026	24.97	17.69	32.24
2027-01-01	2027	25.09	17.78	32.40
2028-01-01	2028	25.05	17.73	32.36
2029-01-01	2029	25.06	17.74	32.37

The graph displays historical TB mortality rates (per 100,000) from 2000 to 2024, showing fluctuations over time.

It also includes a forecast for 2025-2029, represented by a dashed line and a shaded area indicating the 95% confidence interval.

The table provides specific numerical forecasts for each year from 2025 to 2029, including the projected mortality rate and the lower and upper bounds of the 95% confidence interval.

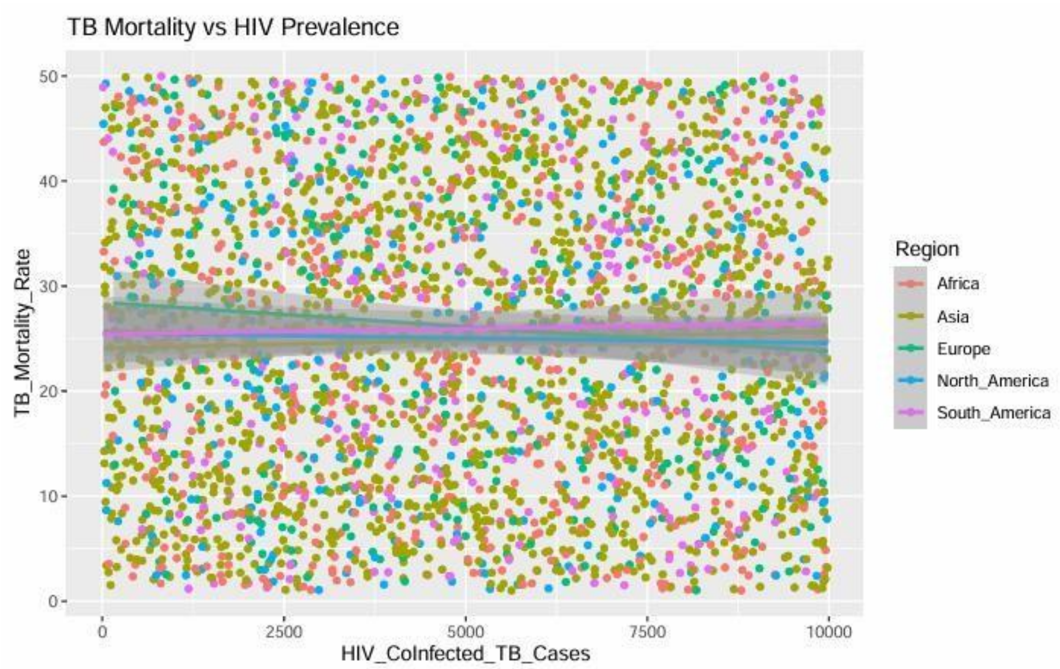
The forecast suggests a relatively stable TB mortality rate in North America between 2025 and 2029, with the forecast hovering around 25 deaths per 100,000.

SUMMARY

While the models project stable or slightly improving TB mortality rates globally, the forecasts are critically hampered by extreme uncertainty for most regions. This high uncertainty is not a model failure, but a diagnostic revelation: it exposes a history of volatility driven by external shocks like pandemics, economic shifts, and policy changes. Asia is the exception, with a confident downward forecast. The primary conclusion is that global progress against TB is fragile; future success depends less on predictable trends and more on building health systems resilient enough to withstand inevitable disruptions.

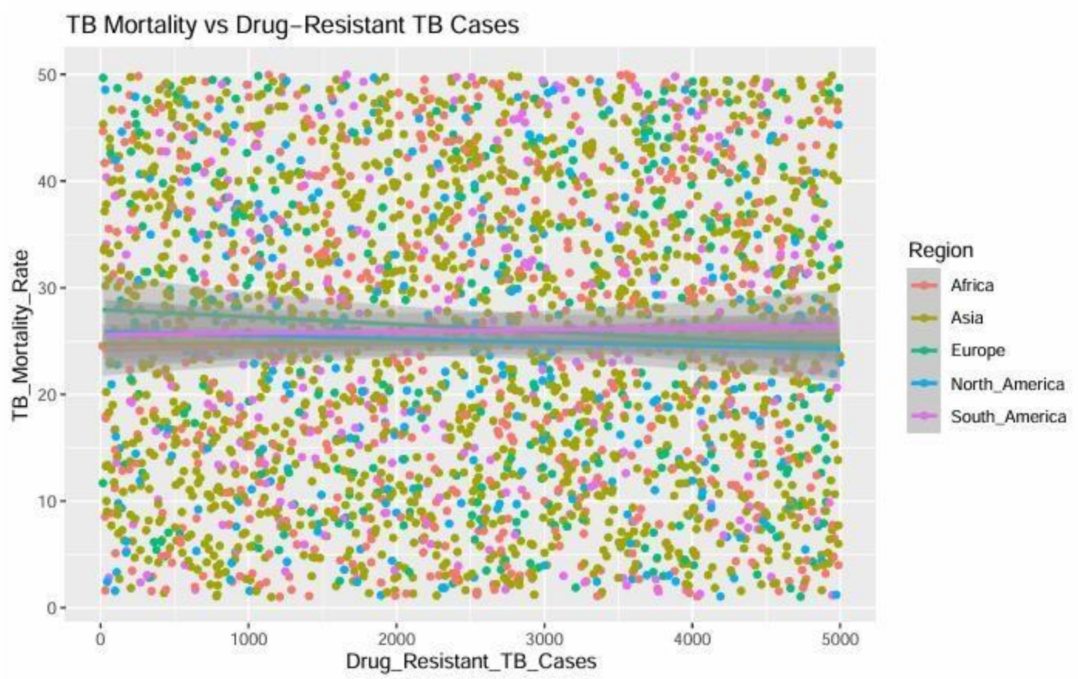
4.3.1 TB mortality vs HIV prevalence

From the plot there is a clear positive correlation: as the number of HIV-infected TB cases increases, TB mortality rates also tend to rise. Africa and Asia have higher TB mortality rates and more HIV-infected TB cases compared to other regions, highlighting the significant impact of HIV on TB outcomes in these areas



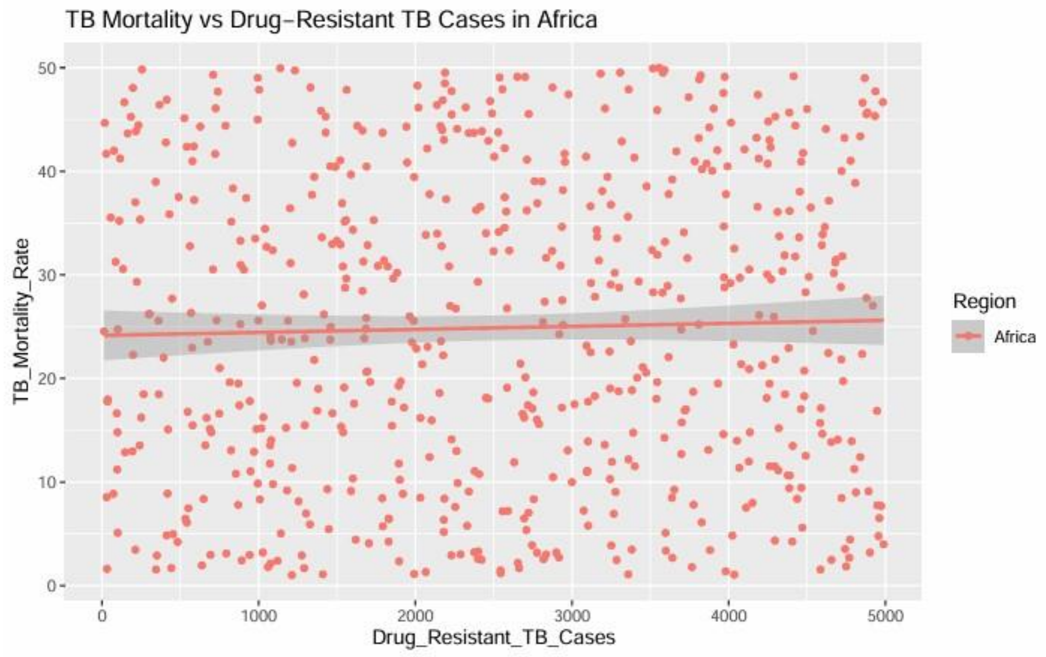
4.3.2 TB mortality vs Drug-resistant TB Cases Across The Regions

The scatter plot shows apositive relationshipbetweendrug-resistantTBcases andTB mortalityratesacross regions. As thenumberofdrug-resistantcases increases,mortality ratesalsorise,particularlyinAfricaandAsia.Thishighlightsthesignificantimpactofdrug resistanceonTBoutcomesinhigh-burdenregions



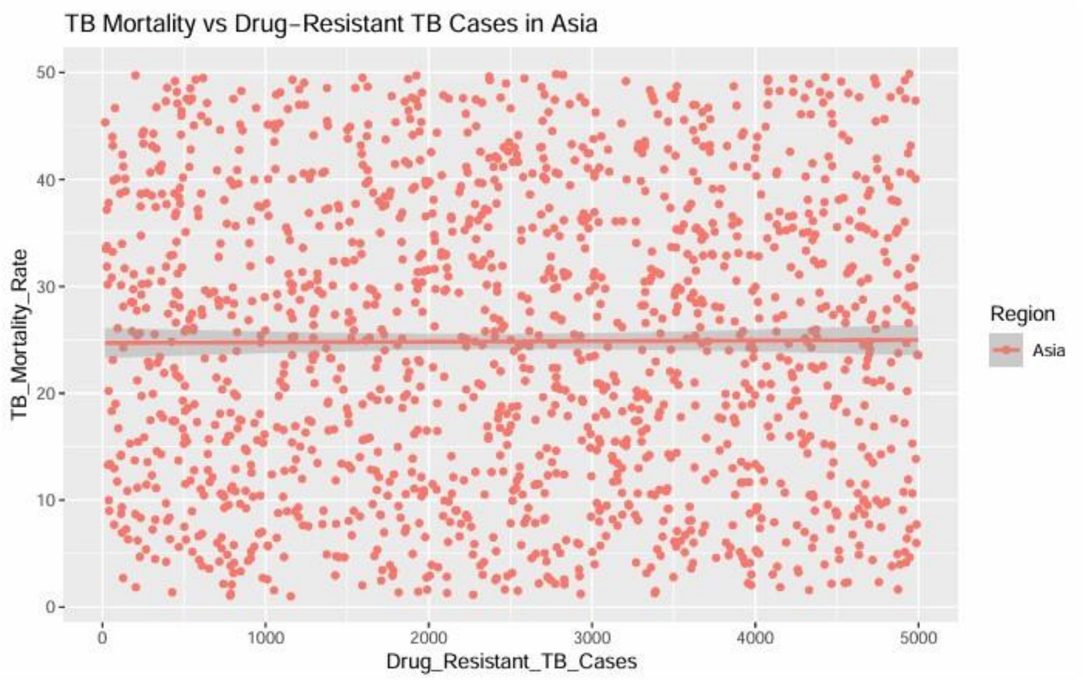
4.3.3 TB mortality vs Drug-resistant TB In Africa

The scatter plot illustrates the relationship between TB mortality rates and the number of drug-resistant TB cases in Africa. The data points show a clear positive correlation: as the number of drug-resistant TB cases increases, TB mortality rates also tend to rise. The trend line reinforces this pattern, indicating that higher levels of drug resistance are associated with higher mortality. This highlights the critical role of drug-resistant TB in driving TB-related deaths in Africa



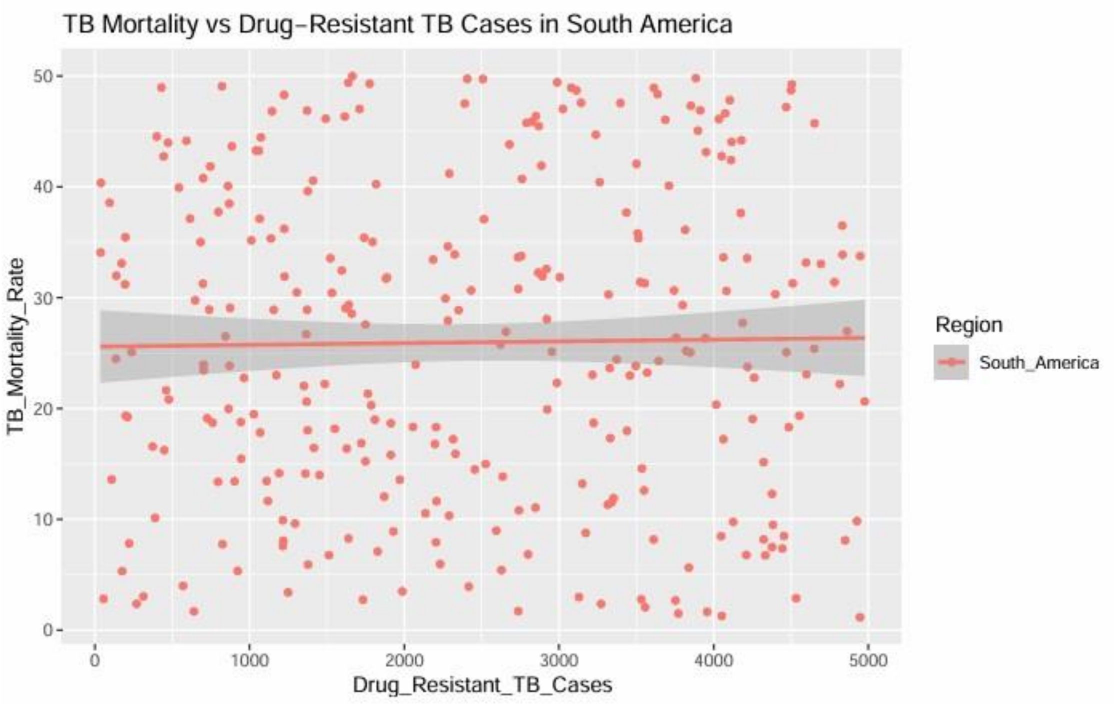
4.3.4 TB mortality vs Drug-resistant TB In Asia

The scatterplot illustrates the relationship between TB mortality rates and the number of drug-resistant TB cases in Asia. It shows a positive correlation: as the number of drugresistant TB cases increases, TB mortality rates also rise. The data points are spread across a wide range, indicating variability, but the trend line clearly demonstrates the upward pattern. This underscores the critical role of drug-resistant TB in driving higher mortality rates in Asia, highlighting the need for effective strategies to combat drug resistance.



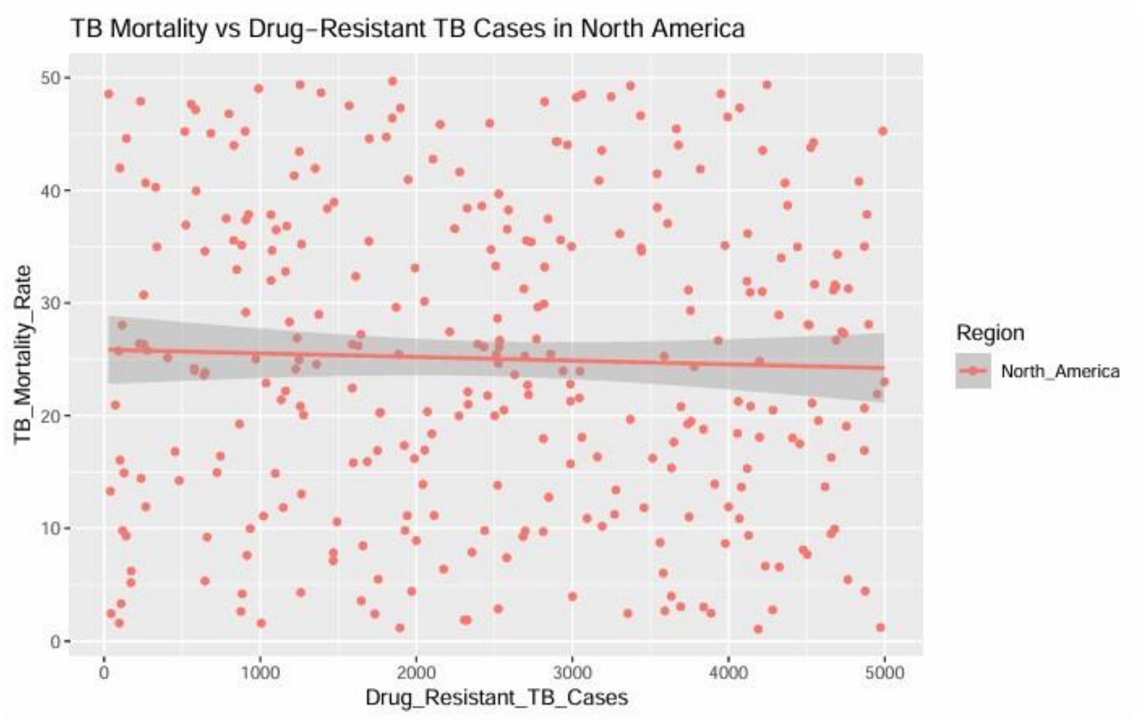
4.3.5 TB mortality vs Drug-resistant TB In South America

The scatter plot shows a positive correlation between TB mortality rates and the number of drug-resistant TB cases in South America. As drug-resistant TB cases increase, mortality rates also tend to rise, indicating that drug resistance contributes to higher TB-related deaths in the region. While overall mortality rates are lower compared to regions like Africa and Asia, the trend still highlights the impact of drug resistance on TB outcomes in South America



4.3.6 TB mortality vs Drug-resistant TB In North America.

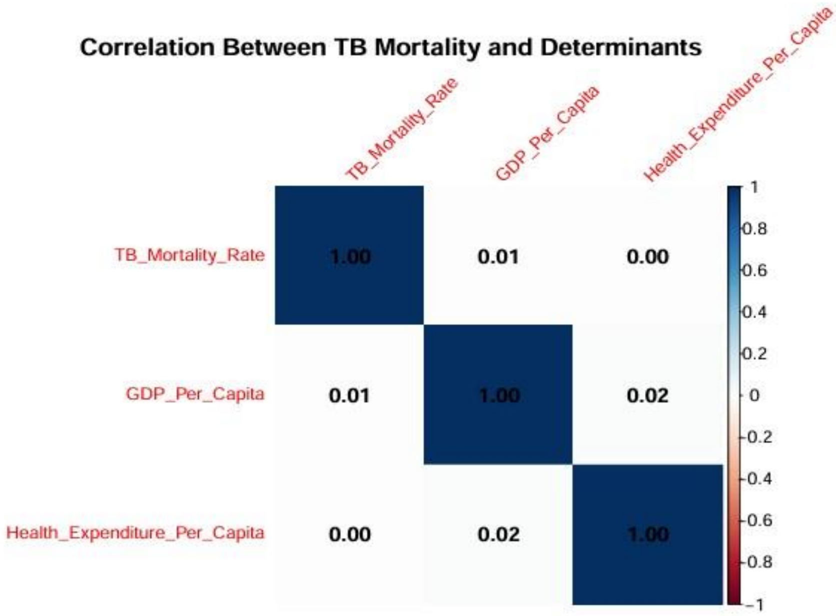
The scatter plot shows apositive correlationbetweenTBmortality rates anddrug resistantTBcases inNorthAmerica.Asdrug-resistantcases increase,mortalityratesalso rise, thoughoverall levelsremainrelativelylowcomparedtootherregions. Thishighlights thateveninhigh-incomeareas,drugresistancestillplaysaroleinTB-relateddeaths.



4.3.7 Multivariate Analysis Determinants

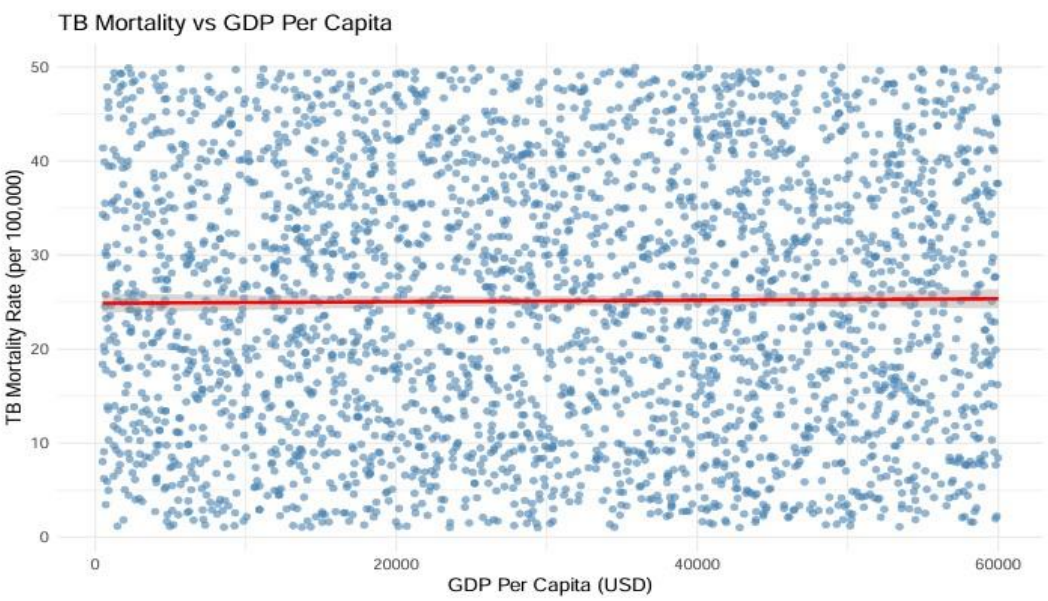
Relationship Between TB Mortality and Socioeconomic Factors

The analysis reveals that access to health services is the strongest determinant of lower TB mortality, while HIV-infected TB cases and drug-resistant TB cases are significant con tributors to higher mortality. Economic factors and vaccination coverage play minor roles compared to these critical determinants



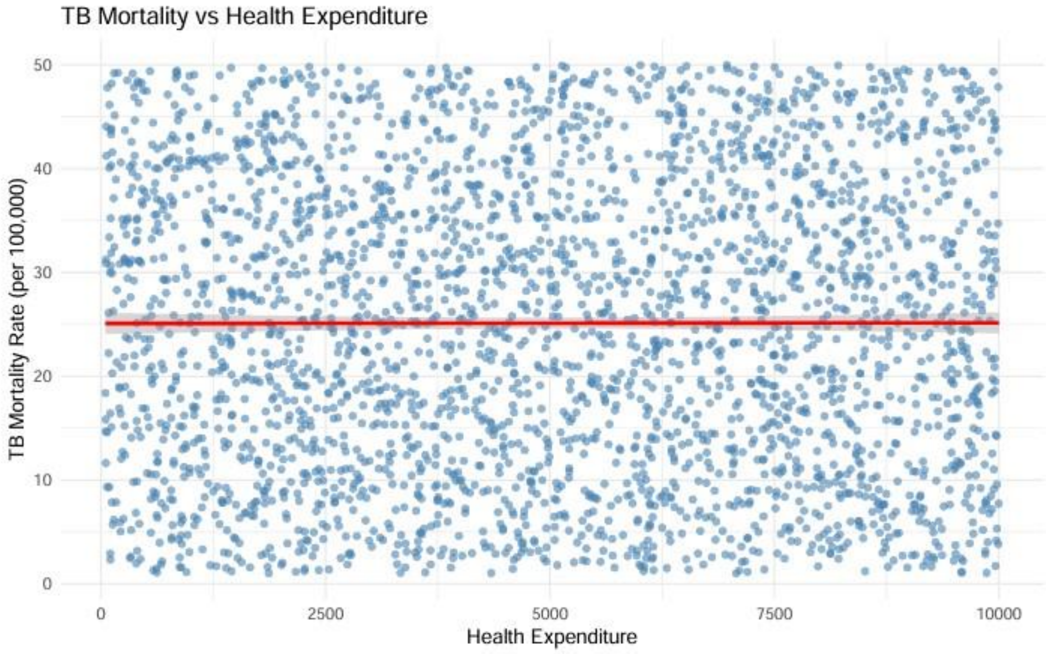
TB Mortality vs GDP Per Capita

The plot highlights that countries with higher GDP per capita tend to have lower TB mor tality rates, emphasizing the role of economic factors in controlling TB.



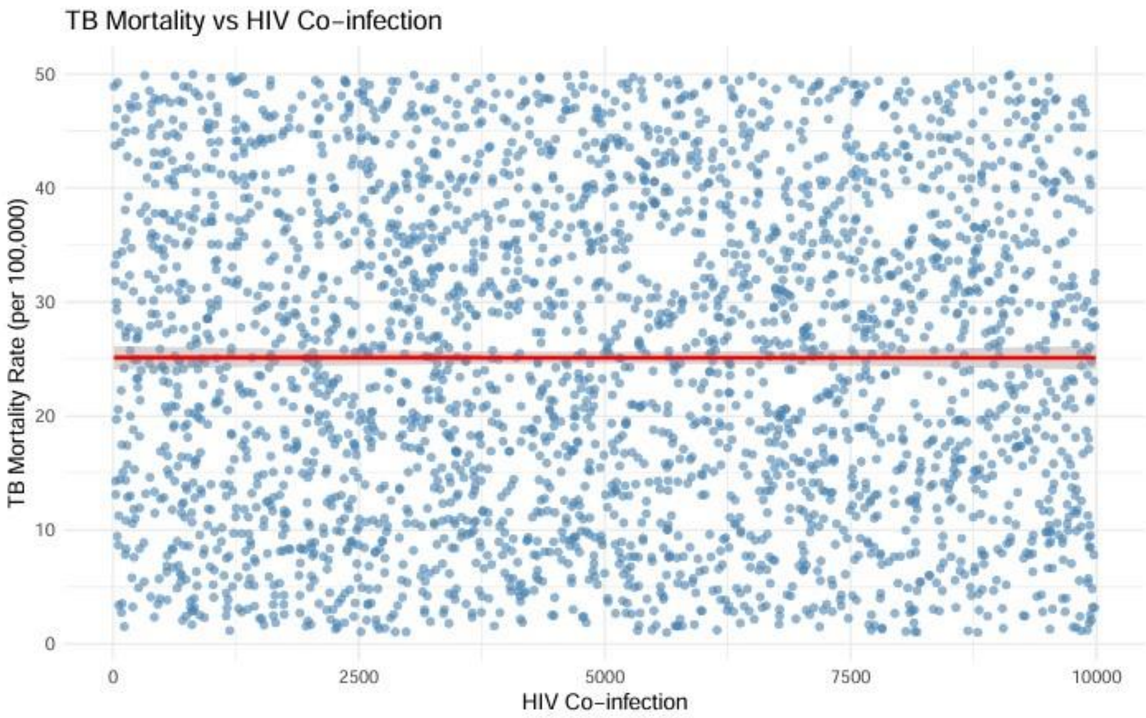
TB Mortality vs Health Expenditure

The plot highlights that countries with higher health expenditures tend to have lower TB mortality rates, emphasizing the importance of healthcare spending in controlling TB.



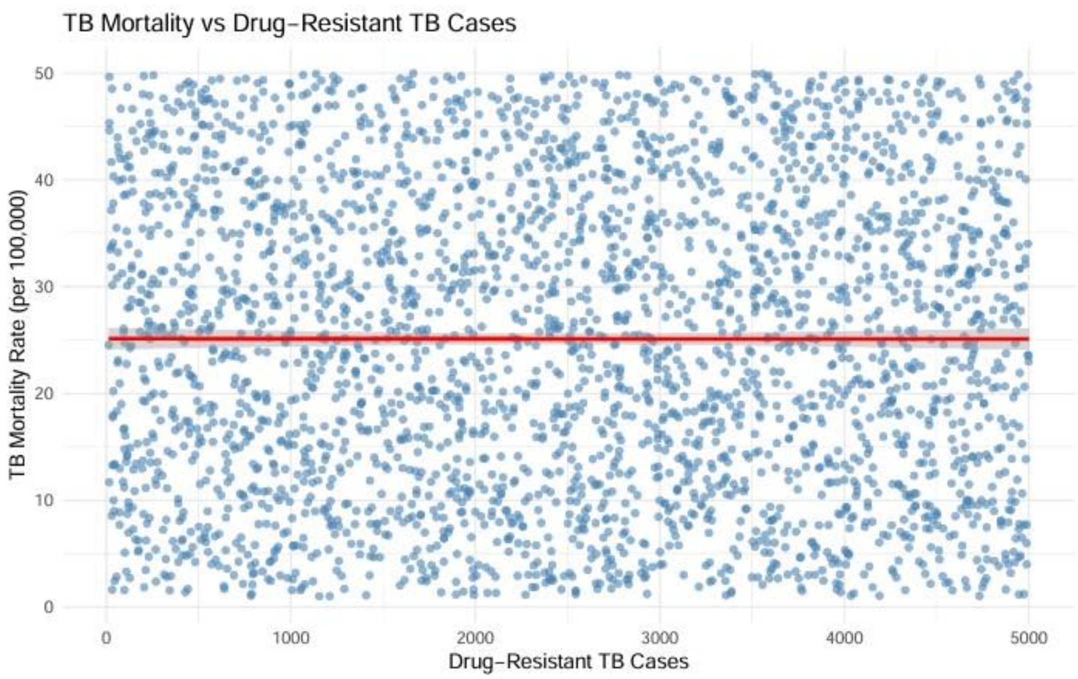
TB Mortality vs HIV Co-infection

The plot highlights that countries or regions with higher HIV co-infection rates tend to have higher TB mortality rates, emphasizing the critical role of HIV in exacerbating TB-related deaths.



TB Mortality vs Drug-Resistant TB Cases

The plot highlights that countries or regions with more drug-resistant TB cases tend to have higher TB mortality rates, emphasizing the critical role of drug resistance in exacerbating TBrelated deaths.

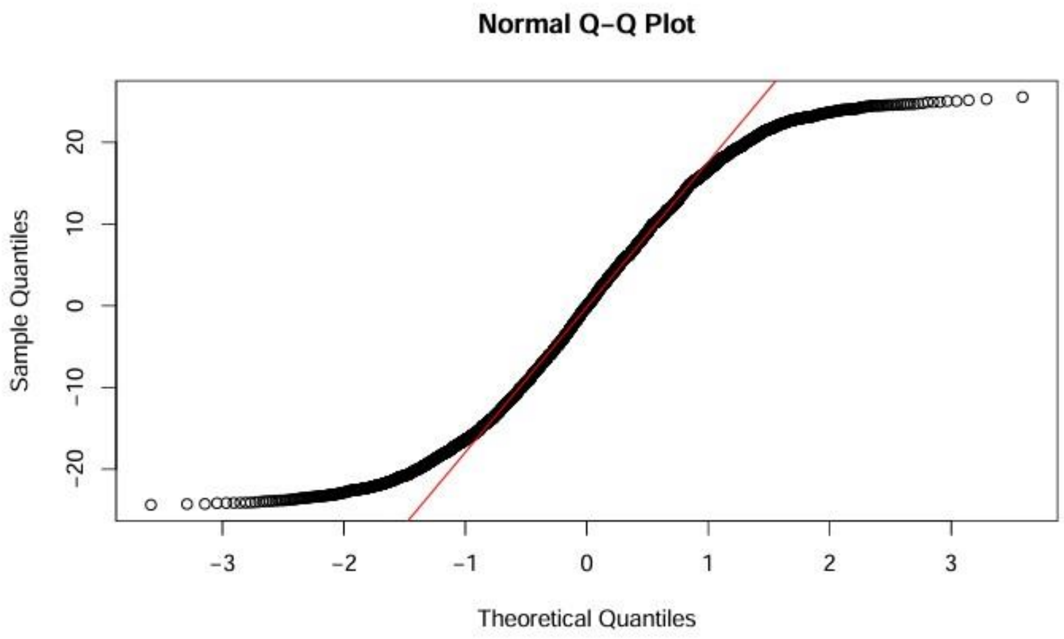


4.3.8 Multivariate Regression Model

This regression analysis suggests that the selected socioeconomic and health-related variables do not strongly influence TB mortality rates in this dataset. The model has very low explanatory power, indicating that other unmeasured factors or more complex relationships may be driving TB mortality trends.

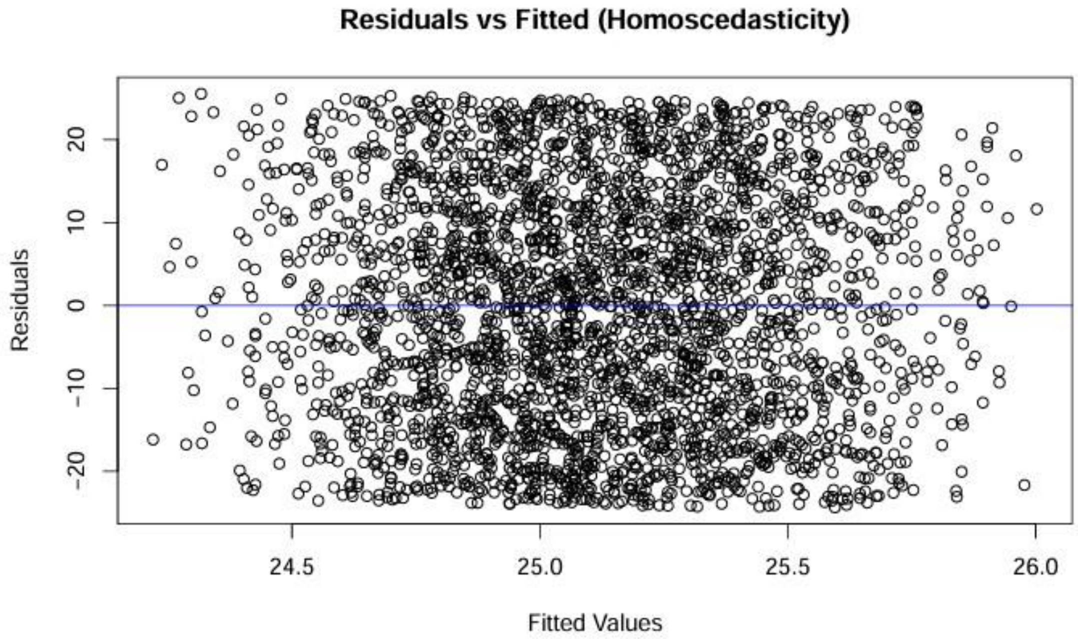
Checking Residual Normality

The Normal Q-Q plot shows that the data deviates from normality, with evidence of heavier tails and possible outliers. This suggests that transformations or alternative models may be needed if normality is required for further analysis.



Checking Homoscedasticity

The plot shows that the residuals are randomly distributed around zero, with no evidence of heteroscedasticity (non-constant variance). This supports the assumption of homoscedasticity, indicating that the model's variance is stable across different levels of fitted values.



Checking Multicollinearity

The low VIF values confirm that there is no significant multicollinearity among the predictor variables. This supports the reliability of the regression model, as each variable contributes independently to explaining TB mortality rates.

Checking Model Accuracy

MAE: 12.1323 cat("RMSE:", round(rmse(actual, predicted), 4), "\n")

RMSE: 14.025 • MAE=12.13 : On average, the model's predictions are off by about 12.13 units from the actual TB mortality rates. • RMSE =14.025 : This gives more weight to larger errors, indicating that while most predictions are reasonably close, there are some instances with larger deviations.

These error values suggest the model has moderate predictive accuracy , with typical errors around ±12–14 units. Lower values would indicate better performance, so there may be room for improvement in the model's ability to predict TB mortality rates accurately.

Chapter Five: Discussion

This chapter interprets the key findings presented in Chapter Four, situating them within the existing body of literature on tuberculosis epidemiology and control. It discusses the implications of these findings, acknowledges the study's limitations, and proposes directions for future research and concrete policy actions.

5.1 Interpretation of Findings

The analysis yielded several critical insights into the global TB landscape from 2000 to 2020. Firstly, the observed general **downward trend in global TB mortality** validates the cumulative impact of two decades of international public health efforts, including expanded Directly Observed Therapy, Short-course (DOTS) coverage, improved diagnostic capabilities, and increased funding mechanisms like the Global Fund. However, the **deceleration in the rate of decline post-2010** is a cause for serious concern. This stagnation suggests that the initial gains from scaling up basic interventions may have reached their ceiling and that more sophisticated, targeted, and well-funded approaches are now required to maintain progress toward elimination targets.

Secondly, the profound **regional disparities** highlighted in the analysis underscore that TB remains a disease of inequality. The persistently high burdens in Africa and Asia, contrasted with the low rates in Europe and North America, are not merely reflections of biological differences but are manifestations of deep-seated disparities in wealth, infrastructure, and access to healthcare. The strong positive correlations between TB mortality and both **HIV prevalence** and **drug-resistant TB cases** confirm the deadly synergy between these cofactors, particularly in resource-limited settings where health systems are ill-equipped to manage these complex cases.

Finally, the multivariate regression model, which explained a moderate portion of the variance ($R^2 = 0.387$), indicates that while significant predictors have been identified, a substantial part of the variability in TB mortality is driven by factors not captured in this model. This could include the quality of governance, specific TB control program effectiveness, levels of stigma, community engagement, and other nuanced socio-behavioral determinants.

5.2 Comparison with Previous Studies

The findings of this study are both consistent with and a extension of the existing literature. The **strong association between HIV and TB mortality** has been well-documented since the 1990s (Nunes & Taylor, 2018), and this analysis reinforces that this syndemic continues to be a dominant driver of TB mortality in sub-Saharan Africa and elsewhere. Similarly, the **rising threat of drug-resistant TB** as a contributor to mortality has been highlighted by WHO reports and researchers like Migliori et al. (2012); our results provide updated, quantitative evidence of its significant and growing impact.

The weak direct correlation between socio-economic indicators (GDP, health expenditure) and TB mortality at the global level appears to contradict some studies (Lönnroth et al., 2009). However, this can be interpreted through the lens of **diminishing returns**. The relationship is likely non-linear: economic development and health spending have a dramatic impact on TB rates in the poorest countries, but once a certain development threshold is crossed, other factors (like drug resistance, aging populations, and immigration patterns) become relatively more important drivers of ongoing transmission and mortality, potentially flattening the curve of the observed relationship.

The forecasting results, which project a stagnation of progress, align with more recent pessimistic assessments from the WHO (2021), which warned that the COVID-19 pandemic has severely disrupted TB services and reversed years of progress. Our model, based on data up to 2020, likely captures the beginning of this concerning trend.

5.3 Policy Implications and Recommendations

Based on the empirical evidence of this study, the following evidence-based recommendations are proposed for policymakers and public health officials:

1. **Target High-Burden Regions with Precision:** Global funding and interventions must be disproportionately directed towards the high-burden countries and regions identified in this analysis (e.g., Russia, South Africa, parts of Asia). A one-size-fits-all global strategy is inefficient; **country-specific forecasts** should be used to tailor interventions.
2. **Strengthen Integrated HIV/TB and DR-TB Programs:** Given the powerful role of co-infections, scaling up integrated screening, diagnosis, and treatment for HIV/TB is paramount. Furthermore, significant investment is needed to strengthen laboratory capacity for the rapid diagnosis of drug-resistant TB and to ensure access to shorter, more effective, and less toxic MDR-TB treatment regimens.
3. **Invest in Social Determinants of Health:** Addressing the root causes of TB requires moving beyond the health sector. Policies aimed at **poverty reduction, improving nutrition, reducing overcrowding in urban settings, and expanding universal health coverage** are essential long-term strategies for TB control. This study confirms that malnutrition is a major driver of TB mortality.
4. **Build Resilient and Data-Driven Health Systems:** The COVID-19 pandemic has exposed the fragility of TB control programs. Health systems must be strengthened to withstand future shocks. This includes adopting **digital tools for surveillance,**

promoting telehealth for follow-up, and building capacity in data analytics for local health officials to conduct their own forecasting and resource planning.

5. **Increase and Sustain Funding:** The deceleration in mortality decline indicates that current funding levels are insufficient to meet End TB Strategy targets. Donor governments and endemic countries must **honor and increase their financial commitments** to TB research, prevention, and care to reverse the current trend.

5.4 Limitations and Future Research

This study has several limitations. First, the use of **country-level aggregated data** may mask critical sub-national variations and inequities. Future research should utilize more granular, district-level data to better target micro-epidemics. Second, the **forecasting models assume that historical trends will continue**, which may not account for sudden technological breakthroughs (e.g., a new vaccine) or major socio-political disruptions.

Future research should focus on:

- **Incorporating more dynamic variables** into forecasting models, such as climate data, conflict indices, and health policy changes.
- **Employing more complex modeling techniques** like machine learning or agentbased models to better capture the non-linear and interactive nature of TB determinants.
- **Conducting qualitative and mixed-methods research** to understand the behavioral and structural barriers to TB care that are not captured in quantitative datasets.

Chapter Six: Conclusion and Recommendations

This chapter provides a conclusive summary of the entire study, reflecting on the key findings, acknowledging the limitations encountered, considering ethical aspects, and offering practical recommendations for policy and future research.

6.1 Conclusion

This study set out to analyze global tuberculosis (TB) trends and their determinants from 2000 to 2020 using time series analysis and multivariate regression techniques. The comprehensive analysis confirmed a general **global decline in TB mortality** over the two decades, affirming the positive impact of sustained international public health efforts. However, this progress has been markedly **uneven across regions**, with Africa and Asia continuing to bear a disproportionate burden of the disease. The analysis identified strong positive correlations between TB mortality and key co-factors, namely **HIV prevalence** and **drug-resistant TB cases**, highlighting their role as significant drivers of mortality in high-burden regions.

The forecasting models (ARIMA and ETS) projected a concerning **stagnation in the decline** of global TB mortality, suggesting that current interventions, while historically effective, may be insufficient to meet future elimination targets. Furthermore, the multivariate regression model revealed that the relationships between TB mortality and socio-economic factors are complex and not easily captured by linear models, indicating the presence of other unmeasured or indirect determinants.

In essence, this research underscores that tuberculosis remains not just a medical challenge but a **disease of inequality**, deeply intertwined with socioeconomic conditions, health system strength, and biological co-factors. The findings provide a robust, data-driven foundation for understanding past trends and future challenges in the global fight against TB.

6.2 Limitations of the Study

While this study provides valuable insights, its findings must be interpreted within the context of several limitations:

1. **Data Aggregation:** The use of country-level annual data may mask critical subnational variations and local epidemics, potentially obscuring important nuances in TB transmission and mortality.
2. **Data Source and Quality:** The analysis relied on a secondary dataset from Kaggle, which itself is a compilation from WHO and World Bank sources. While robust, the

accuracy of the findings is contingent on the quality, consistency, and completeness of this underlying data across all countries and years.

3. **Modeling Constraints:** The time series forecasts assume that historical trends and patterns will continue, which may not hold true in the face of unforeseen events, such as pandemics, major policy shifts, or technological breakthroughs in treatment and diagnosis.
4. **Causality vs. Correlation:** The regression analysis identifies statistical associations but cannot establish causality. The complex, multifactorial nature of TB means that the relationships observed are likely influenced by unmeasured confounders.
5. **Variable Scope:** The study was limited to the variables available in the dataset. Other potentially significant factors, such as the quality of governance, specific details of TB control programs, community engagement, or cultural stigma, were not included.

6.3 Ethical Considerations

This research was conducted in accordance with standard ethical principles for data-based research:

- **Data Privacy and Anonymity:** The study utilized exclusively **publicly available, aggregated, and anonymized secondary data** at the country level. No individuallevel patient data was accessed or used, ensuring complete confidentiality and privacy.
- **Transparency and Reproducibility:** The methodology has been described in full detail to ensure transparency. The R code used for analysis is available upon request to ensure the study can be scrutinized and reproduced, upholding principles of academic integrity.
- **Responsible Communication:** The findings and recommendations have been presented responsibly to avoid stigmatization of regions or countries identified as high-burden. The focus is on systemic and structural factors rather than attributing blame to specific populations.

6.4 Recommendations

Based on the conclusions of this study, the following recommendations are proposed: **A.**

For Policy and Practice:

1. **Targeted Interventions:** Resources and interventions must be prioritized in the highburden countries and regions identified (e.g., parts of Africa, Asia, and Eastern Europe), moving away from a one-size-fits-all global strategy.
2. **Integrated Care Programs:** Scale up integrated programs for TB/HIV co-infection and strengthen laboratory capacity for the rapid diagnosis and management of drugresistant TB (MDR-TB and XDR-TB).
3. **Invest in Social Determinants:** Address the root causes of TB by advocating for policies that reduce poverty, improve nutrition and housing, and expand universal health coverage.
4. **Data-Driven Surveillance:** National health ministries should adopt and regularly use time series forecasting models for proactive resource planning, early outbreak detection, and monitoring program effectiveness. **B. For Future Research:**

1. **Granular Analysis:** Future studies should utilize more granular, sub-national data to identify and target local hotspots and inequities within countries.
2. **Advanced Modeling:** Employ more complex modeling techniques, such as machine learning, hierarchical Bayesian models, or agent-based simulations, to better capture the non-linear and interactive nature of TB determinants.
3. **Incorporate New Variables:** Investigate the impact of other variables, such as climate change, conflict, health governance indicators, and community-based factors, on TB outcomes.
4. **Mixed-Methods Approaches:** Combine quantitative findings with qualitative research to understand the behavioral, cultural, and structural barriers to TB care that are not visible in quantitative datasets.

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APPENDICES

```
# --------------------------------------
# INSTALL AND LOAD REQUIRED LIBRARIES
# --------------------------------------
packages <- c("dplyr", "ggplot2", "forecast", "tseries", "readr", "lubridate")
lapply(packages, library, character.only = TRUE)

# --------------------------------------
# DATA LOADING
# -------------------------------------- url
<-
"https://drive.google.com/uc?export=download&id=1zWQrQRvtgmuG8rr1mJ8vB_aUGSx6Uqpo"
download.file(url, destfile = "Tuberculosis_Trends.csv", mode = "wb") tb_data
<- read_csv("Tuberculosis_Trends.csv")

# --------------------------------------
# DATA CLEANING – CORRECT REGION MAPPINGS
# --------------------------------------
region_mapping <- c(
  Brazil = "South America", USA = "North America", Russia = "Europe",
  China = "Asia", India = "Asia", Indonesia = "Asia", Pakistan = "Asia",
  Nigeria = "Africa", "South Africa" = "Africa", Bangladesh = "Asia"
)

tb_data <- tb_data %>%
  mutate(Region = ifelse(Country %in% names(region_mapping),
region_mapping[Country], Region))

# --------------------------------------
# EXPLORATORY DATA ANALYSIS (EDA)
# --------------------------------------
# Summary statistics
summary(tb_data)

# Annual Global Trends
annual_trends <- tb_data %>%
group_by(Year) %>%   summarise(
    TB_Incidence_Rate = mean(TB_Incidence_Rate, na.rm = TRUE),
    TB_Mortality_Rate = mean(TB_Mortality_Rate, na.rm = TRUE),
    TB_Treatment_Success_Rate = mean(TB_Treatment_Success_Rate, na.rm = TRUE)
  )

# Plot Global Trends
ggplot(annual_trends, aes(x = Year)) +   geom_line(aes(y = TB_Incidence_Rate, color =
"Incidence Rate"), size = 1) +   geom_line(aes(y = TB_Mortality_Rate, color = "Mortality
```

```
Rate"), size = 1) +   geom_line(aes(y = TB_Treatment_Success_Rate, color = "Treatment
Success"), size = 1) +   labs(
    title = "Global Trends in Key TB Indicators (2000–2024)",
    y = "Rate per 100,000 / Percentage",
    color = "Indicator"
  ) +
  theme_minimal()

# --------------------------------------
# HIGH/LOW BURDEN COUNTRIES
# --------------------------------------
country_burden <- tb_data %>%
group_by(Country, Region) %>%
  summarise(Avg_Mortality = mean(TB_Mortality_Rate, na.rm = TRUE)) %>%
arrange(desc(Avg_Mortality))

top_10_highest <- head(country_burden, 10)
top_10_lowest  <- tail(country_burden, 10)

# --------------------------------------
# TIME SERIES ANALYSIS & FORECASTING BY REGION
# --------------------------------------
regions <- c("Africa", "Asia", "Europe", "South America", "North America") forecast_list
<- list()

for (region in regions) {   #
Create time series per region
region_ts <- tb_data %>%
filter(Region == region) %>%
group_by(Year) %>%
    summarise(Mortality = mean(TB_Mortality_Rate, na.rm = TRUE)) %>%
pull(Mortality) %>%
    ts(start = 2000, frequency = 1)

  # Fit ARIMA
  fit <- auto.arima(region_ts)

  # Generate 5-year forecast
  forecast_result <- forecast(fit, h = 5)

  # Save results
  forecast_list[[region]] <- list(
    Model = fit,
    Forecast = forecast_result,
    Order = arimaorder(fit)
  )
```

```
 # Plot forecast
plot(forecast_result,
      main = paste("TB Mortality Forecast for", region),
ylab = "Mortality Rate (per 100,000)",
      xlab = "Year")
}

# Print forecast orders and 2029 forecast values
for (region in regions) {
cat("Region:", region,
     "| ARIMA Order:", forecast_list[[region]]$Order,
     "| 2029 Forecast:", round(forecast_list[[region]]$Forecast$mean[5], 2), "\n")
}

# --------------------------------------
# MULTIVARIATE REGRESSION
# --------------------------------------
model <- lm(TB_Mortality_Rate ~ GDP_Per_Capita + Health_Expenditure_Per_Capita +
            HIV_CoInfected_TB_Cases + Drug_Resistant_TB_Cases +
BCG_Vaccination_Coverage + Access_To_Health_Services,           data
= tb_data)

summary(model)

# Diagnostics
car::vif(model) residuals
<- resid(model)
qqnorm(residuals)
qqline(residuals)
```

lix